



Digital Forensics

Taras Holotyak

Content of course

The purpose of the course is to introduce into the subject together with providing of a practical skills in the domain of digital forensics.

The course will cover the following topics:

- Digital forensics principles
- State of the art of modern multimedia forensics
- Image/video attributing to the source
- Multimedia integrity and authenticity
- Anti-forensics attacks

Course grades

Lectures (CC/Oral exam): 40%	
Labs: 30%	Mini-project (30%)

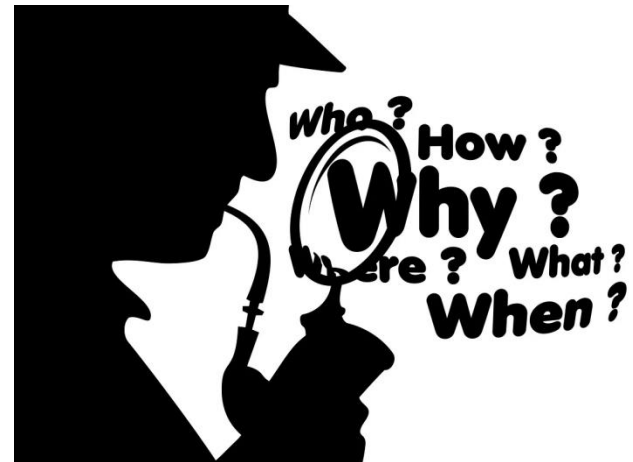
Agenda

- Principles of digital forensics
- Introduction to digital multimedia forensics
- General principles of multimedia forensics
- Examples of media forensic applications

Principles of digital forensics

What is Digital forensics?

- Emerging discipline in computer sciences
 - “voodoo science”
 - still not completely standardized
 - a lot of on-going research
- Investigation that takes place after an incident has happened
- Try to answer questions: who, what, when, where, why, and how?



Principles of digital forensics

Typical investigation steps:

1. Acquisition
2. Identification
3. Evaluation
4. Presentation



Principles of digital forensics

Step 1: Acquisition

Physically or remotely obtaining possession of the computer, all network mappings from the system, and external physical storage devices

- Analogous to crime scene in the “real world”
- Goal is to recover as much evidence without altering the crime scene
- Investigator should document as much as possible
- Create 1:1 copies of evidence

Step 2: Identification

This step involves identifying what data could be recovered and electronically retrieving it by running various digital forensic tools and software utilities

- Goal is to extract data from the acquired evidence
- Always work on copies, never the original
 - must be able to repeat entire process from scratch
- Data, deleted data, “hidden” data: locating hidden or encrypted data is difficult and might even be impossible
- Investigator has to look at other clues:
 - steganography software
 - crypto software



Principles of digital forensics

Step 3: Evaluation

Evaluating the information/data recovered to determine if and how it could be used against the suspect

- Apply corresponding methodologies: methodology differs depending on the objectives of the investigation, e.g., authorship analysis, determination if a system was compromised, etc.
- Utilize system and external information
- Establish time line of events

Step 4: Presentation

This step involves the presentation of evidence discovered in a manner which is understood by lawyers, non-technically staff/management, and suitable as evidence as determined by international and internal laws

- An investigator that performed the analysis may have to appear in court as an expert witness
- For internal investigations, a report or presentation may be required



Principles of digital forensics

What is Digital forensics?

Digital Forensic Science

The use of scientifically derived and proven methods toward the preservation, collection, validation, identification, analysis, interpretation, documentation and presentation of digital evidence derived from digital sources for the purpose of facilitating or furthering the reconstruction of events found to be criminal, or helping to anticipate unauthorized actions shown to be disruptive to planned operations.

Report of the First Digital Forensic Research Workshop (DFRWS) - August 7-8, 2001 - Utica, New York

- Evidence might be required for a wide range of digital crimes and misuses
- Multiple methods of
 - Discovering data on computer system
 - Recovering deleted, encrypted, or damaged file information
 - Monitoring live activity
 - Detecting violations of corporate policy
- Information collected assists in arrests, prosecution, termination of employment, and preventing future illegal activity

Principles of digital forensics

What constitutes digital evidence?

- Any information being subject to human intervention or not, that can be extracted from a digital systems
- Must be in human-readable format or capable of being interpreted by a person with expertise in the subject

Digital forensics examples

- Recovering of deleted emails
- Performing investigation post employment termination
- Recovering evidence post formatting hard drive
- Performing investigation after multiple users had taken over the system



Principles of digital forensics

Who uses Digital forensics?

- Criminal Prosecutors
 - Rely on evidence obtained from a computer to prosecute suspects and use as evidence
- Civil Litigations
 - Personal and business data discovered on a computer can be used in fraud, divorce, harassment, or discrimination cases
- Insurance Companies
 - Evidence discovered on computer can be used to mollify costs (fraud, worker's compensation, arson, etc)
- Private Corporations
 - Obtained evidence from employee computers can be used as evidence in harassment, fraud, and embezzlement cases
- Law Enforcement Officials
 - Rely on computer forensics to backup search warrants and post-seizure handling
- Individual/Private Citizens
 - Obtain the services of professional computer forensic specialists to support claims of harassment, abuse, or wrongful termination from employment

Principles of digital forensics

History of Digital forensics

- Earliest notion of digital forensics came when Federal Rules (US) of evidences first started to discuss digital evidences in the 1970s
- Real digital forensics investigations started in the mid-to late 1980s when federal agents had to start figuring out ways to search computers for digital evidence
- The “home-grown”. Bottom-up approach continue until the late 1990s when security researches at universities and labs started to figure out that this problem was big enough to warrant investigation
- First research groups started around 2000

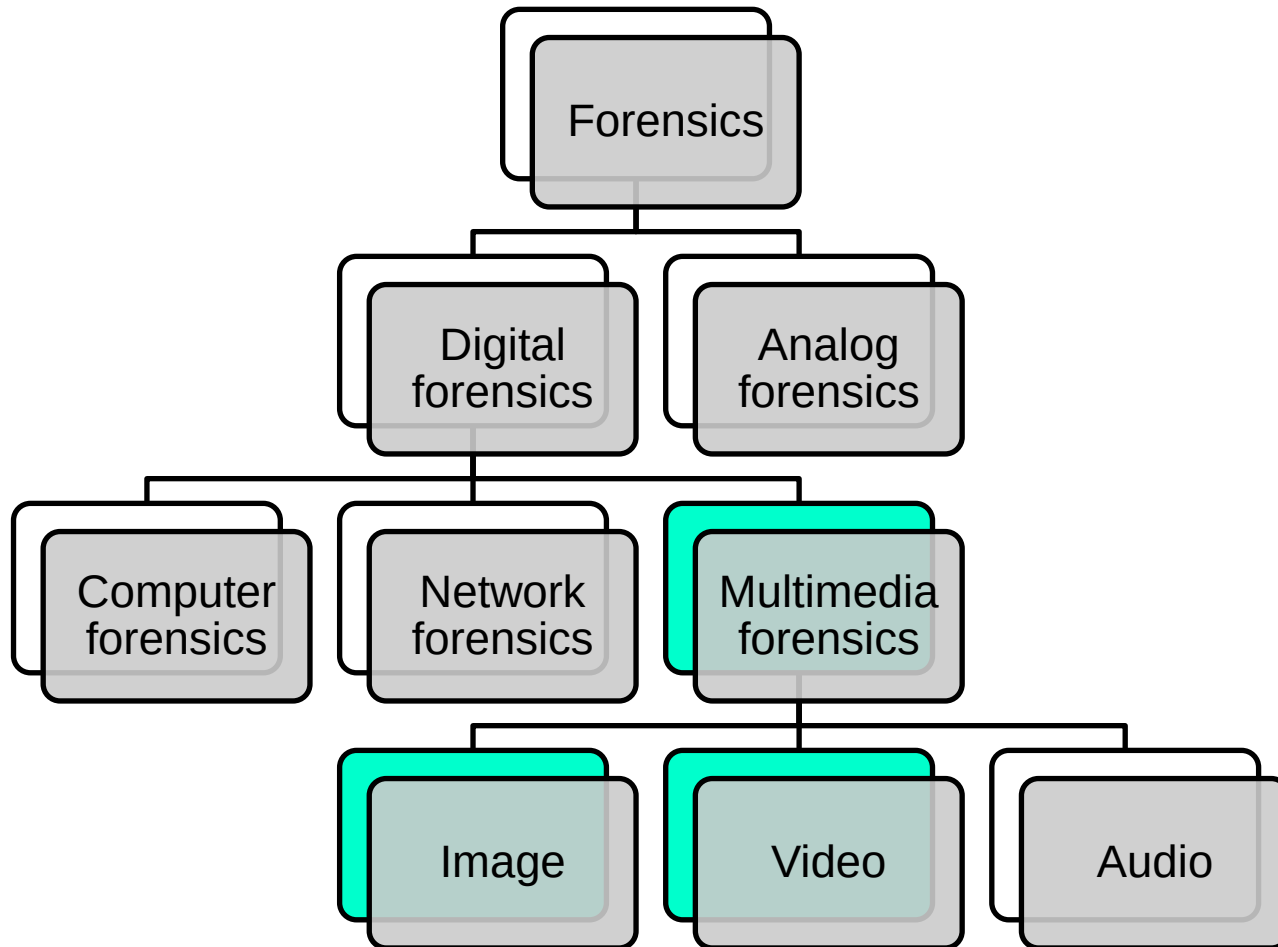
Principles of digital forensics

Digital forensics classes in UNIGE (in scope the direction of cybersecurity)

- Digital Forensics
- Information Systems Security
- Deep Learning
- Sécurité avancée
- Sécurité et confidentialité de multimédia



Principles of digital forensics



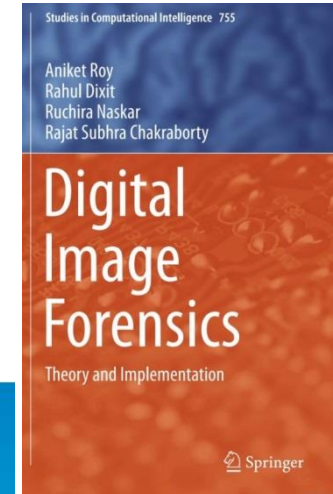
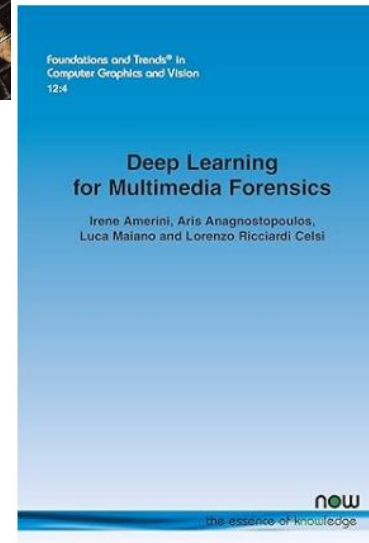
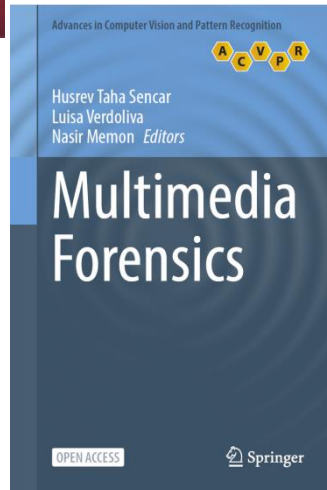
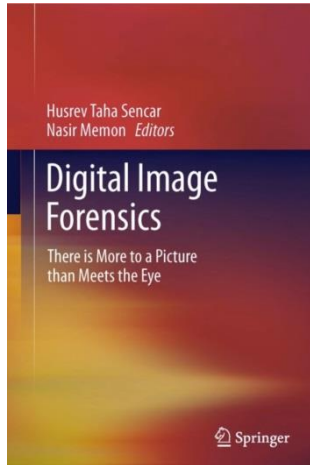
Introduction to digital multimedia forensics

IEEE TRANSACTIONS ON **INFORMATION FORENSICS AND SECURITY**

MMF	MULTIMEDIA FORENSICS
MMF-DEV	Acquisition device forensics
MMF-PROC	Multimedia processing forensics
MMF-PROC-IM	Image forensics
MMF-PROC-VID	Video forensics
MMF-PROC-AU	Audio forensics
MMF-PROC-TXT	Text fingerprinting
MMF-PROC-MM	Multimodal forensic analysis
MMF-MCLUE	Multi-clue forensics analysis
MMF-BENM	Benchmarking
MMF-BENM-REF	Reference datasets and algorithms
MMF-BENM-PER	Performance measurement and validation of forensic evidence
MMF-ADV	Adversarial multimedia forensics
MMF-ADV-COU	Counter forensics
MMF-ADV-ACOU	Anti counter-forensics measures
MMF-ADV-GT	Game theory in multimedia forensics
MMF-OTHS	Others

Introduction to digital multimedia forensics

Recommended books



Introduction to digital multimedia forensics

Photographic images have lost their innocence a long time ago ... (1860)



The world's first **photograph** made by camera was taken in 1826 by Joseph Nicéphore Niépce.

Introduction to digital multimedia forensics

Leon Trotsky, Russian revolutionary



Formerly close comrades, Trotsky appears in the image on the left at one of Lenin's speeches; the same image, altered after the two split, shows Trotsky deleted.

Introduction to digital multimedia forensics

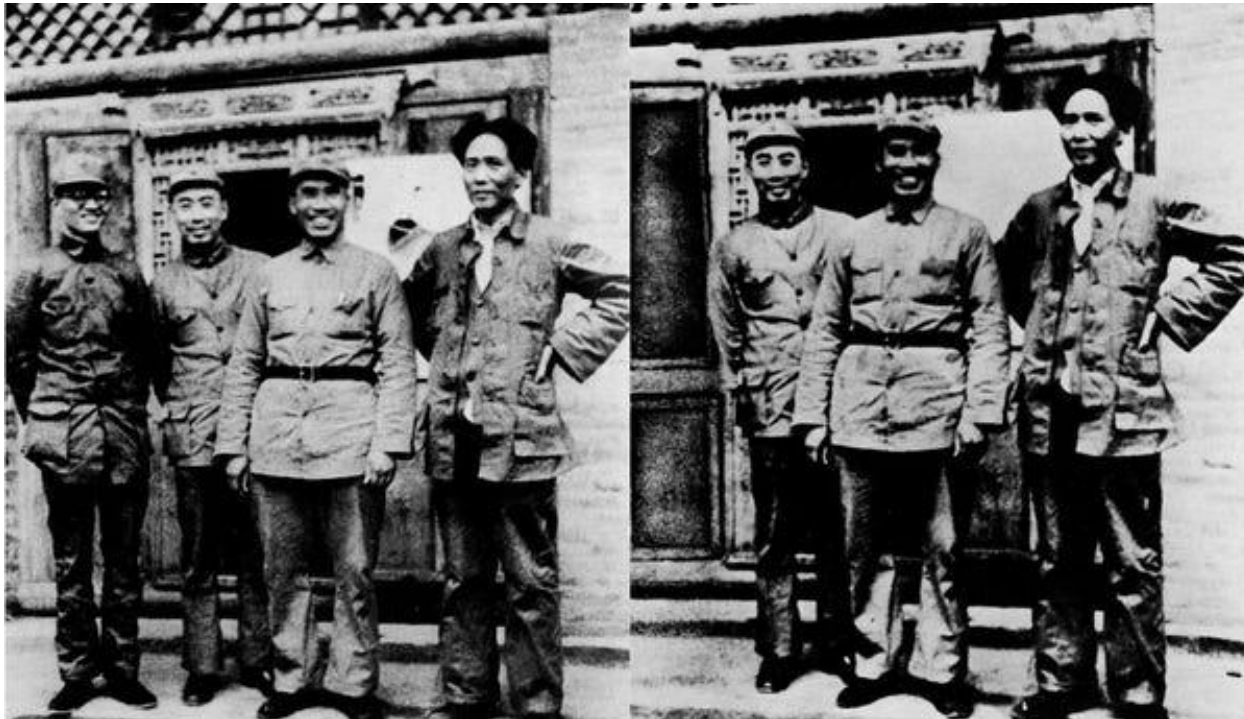
Nikolai Yezhov, Joseph Stalin's head of secret police



Stalin (center) with Nikolai Yezhov to his left. After Yezhov's execution, he was airbrushed out of the photo.

Introduction to digital multimedia forensics

Bo Gu, senior leader of the Chinese Communist Party



Bo Gu, far left, appears in the photo with Mao Zedong and comrades; in the later photo, he is missing.

Introduction to digital multimedia forensics



Jang Song-thaek (Kim Jong Un's uncle and close advisor) disappears in the photo on the right

Introduction to digital multimedia forensics

Seeing is believing ?

With the diffusion of digital images, the validity of photos as witnesses of real events is definitely lost

Introduction to digital multimedia forensics

Celebrities, politics



Introduction to digital multimedia forensics

Frightening enemies



Introduction to digital multimedia forensics

Artificially-augmented support



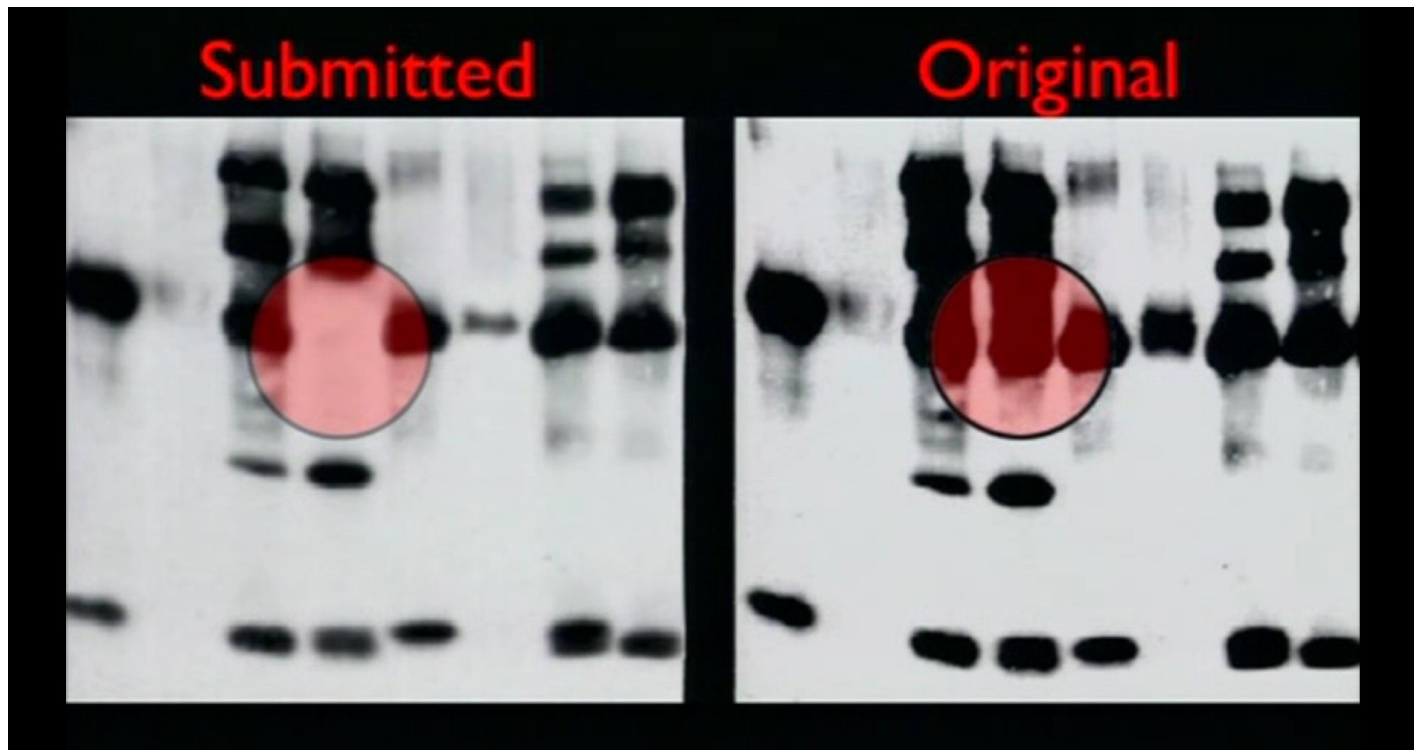
Introduction to digital multimedia forensics

Impressions from Hurricane Sandy



Introduction to digital multimedia forensics

Falsified results of scientific research



Introduction to digital multimedia forensics

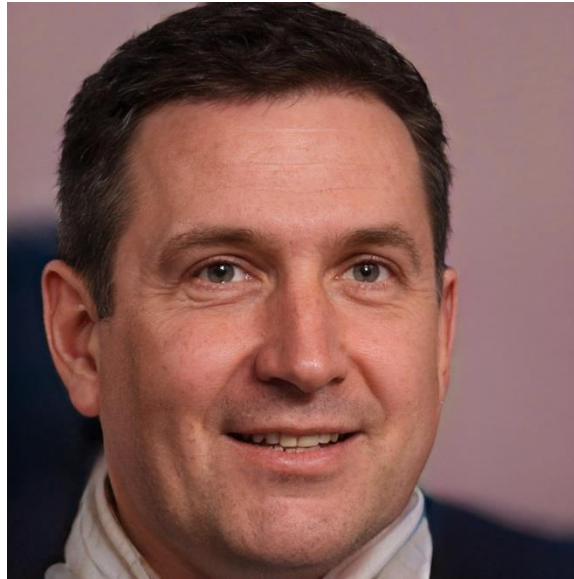
Not only on still images



<https://www.youtube.com/watch?v=CE0Q904gtMI>

Introduction to digital multimedia forensics

Dark side of AI



<https://www.thispersondoesnotexist.com/>

Introduction to digital multimedia forensics

Dark side of AI



winter Yosemite → summer Yosemite



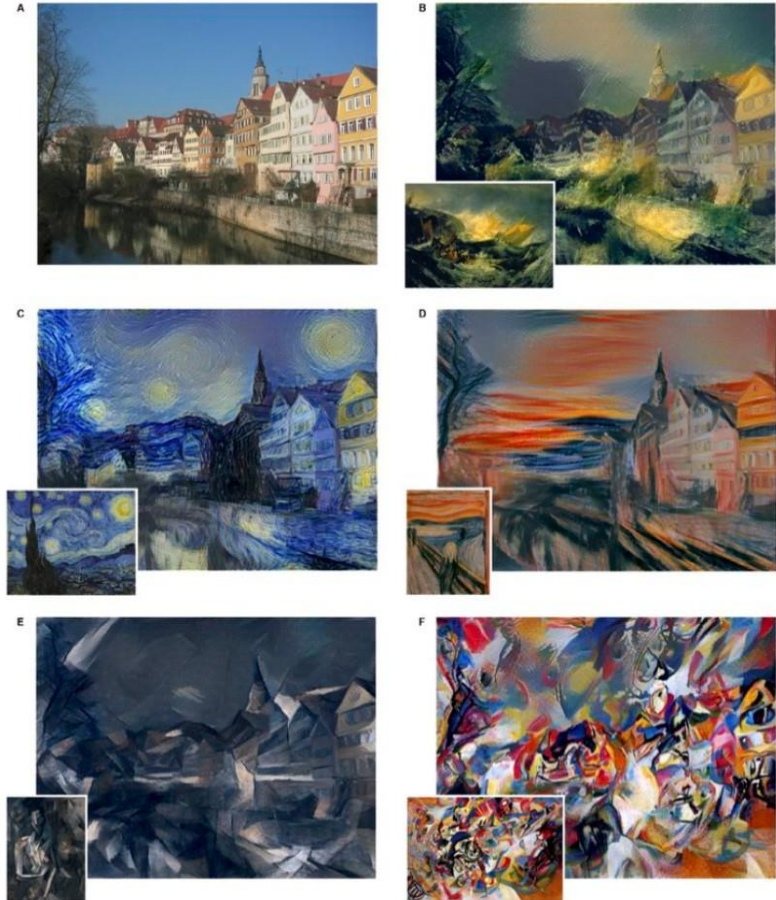
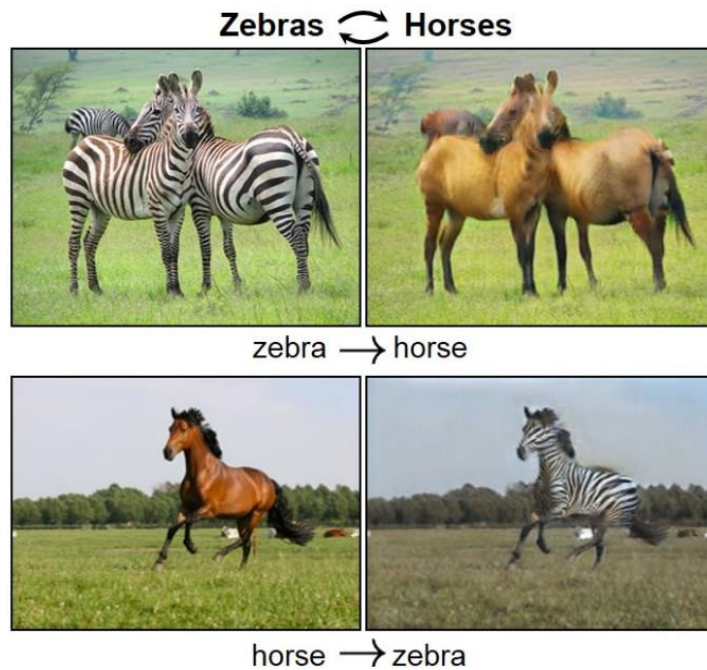
summer Yosemite → winter Yosemite



Style (season) transfer

Introduction to digital multimedia forensics

Dark side of AI



Introduction to digital multimedia forensics

Digital multimedia forensics: why it is important?

- Opinion manipulation
- Social impact: undermines one of our primary source of information
- Unclear value of digital images, videos, audios
- Scientific question: ultimate reliability of digital media as trustful representation of reality

General principles of multimedia forensics

Two approaches to multimedia authentication

- **Active approach:**
 - Cryptographic Signature:
 - Extracting features for generating authentication signature at the source side and verifying the image integrity by signature comparison at the receiver side
 - Possibly coupled with blockchain technology
 - It requires a complete cryptographic infrastructure

Conventional crypto alone leaves many problems unsolved

- Directly apply conventional encryption to compressed multimedia bitstream?
 - May lose error resilience and scalability*
 - Require much computation power*
 - Exploring multimedia property in encryption is desired*
- Distinguish changes by compression vs. malicious tampering?
 - Bit-by-bit accuracy is not always desired authenticity criterion for multimedia*
- Protection from encryption vanishes once data is decrypted
 - Still want establish ownership and restrict illegal re-distributions*

General principles of multimedia forensics

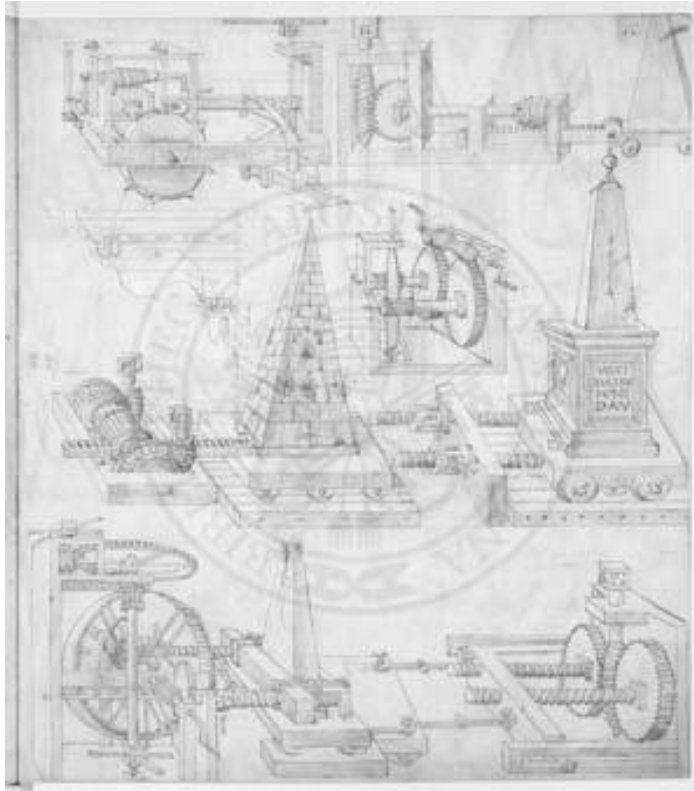
Two approaches to multimedia authentication

- **Active approach:**
 - Fragile/Semi Fragile Digital Watermarking
 - Inserting digital watermark at the source side and using the watermark to verify integrity at the detection side.
 - Two approaches based on:
 - Fragile watermarking
 - Robust watermarking

General principles of multimedia forensics

Visible Digital Watermarks

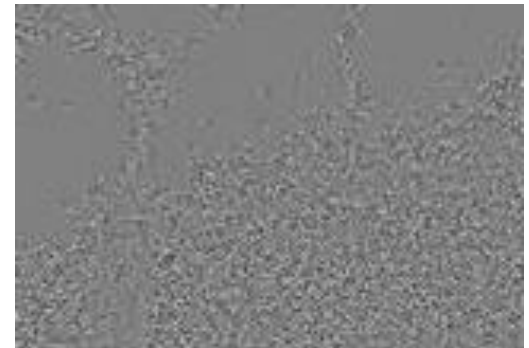
UMCP ENEE739M Slides (created by M. Wu © 2002)



from IBM Watson web page
“Vatican Digital Library”

General principles of multimedia forensics

Invisible Digital Watermarks



1st & 30th Mpeg4.5Mbps frame of original, marked, and their luminance difference (human visual model is used for imperceptibility: protect smooth areas, embed into textures and sharp edges)

General principles of multimedia forensics

Two approaches to multimedia authentication

- **Passive and blind approach: multimedia forensics**
 - Given a digital data (i.e. image), multimedia forensic techniques try to answer a number of forensic questions related to:
 - **source identification**: what is the origin of the data?
 - **integrity verification / tampering detection**: has the data undergone some processing ?
 - Advantages:
 - No need for watermark embedding or signature generation at the source side.
 - No need for a standard
 - No need for a priori knowledge about the acquisition device

General principles of multimedia forensics

Source identification

- How was the image captured?
- Which CLASS of device was used?



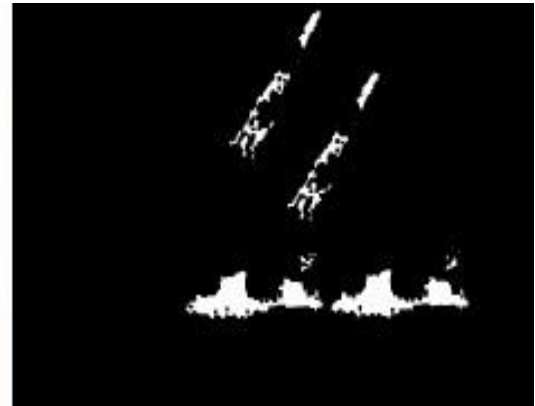
- Which BRAND / MODEL / SPECIFIC DEVICE?



General principles of multimedia forensics

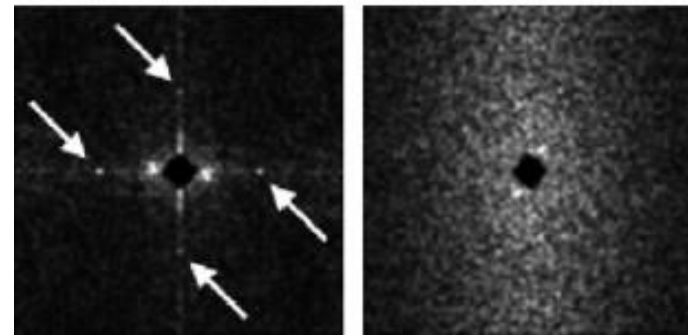
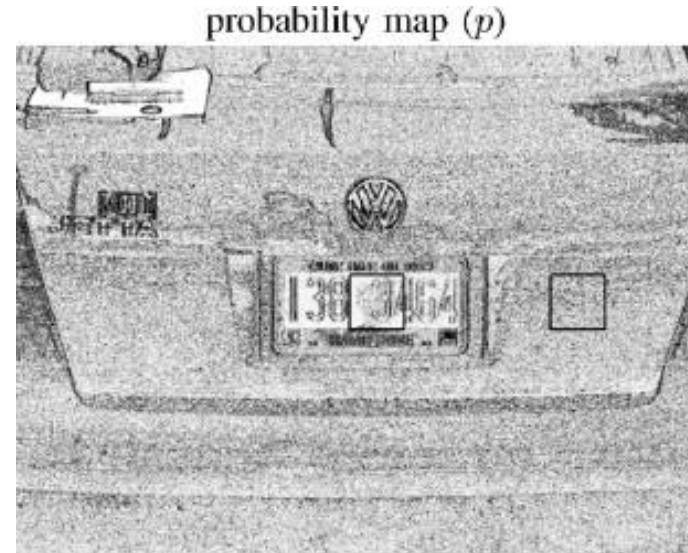
Manipulation detection

- Is the image authentic?
- How was it tampered with?



General principles of multimedia forensics

Manipulation detection



General principles of multimedia forensics

The basic idea

Multimedia forensics is based on the idea that inherent traces (like **digital fingerprints**) are left behind in a digital media during both the creation phase and any other subsequent processing

Digital fingerprints

- **In-camera fingerprints**: each component in the acquisition device leaves intrinsic fingerprints in the final output, due to the specific optical system, color sensor and camera software
- **Out-camera fingerprints**: each processing applied to digital media modifies their properties (e.g. statistical, geometrical, etc.) leaving peculiar traces
- **Scene (geometric) fingerprints**: the real world has specific properties depending on the content, like lighting properties, which characterize the reproduced scene (illuminant direction, specular highlights in the eye)

General principles of multimedia forensics

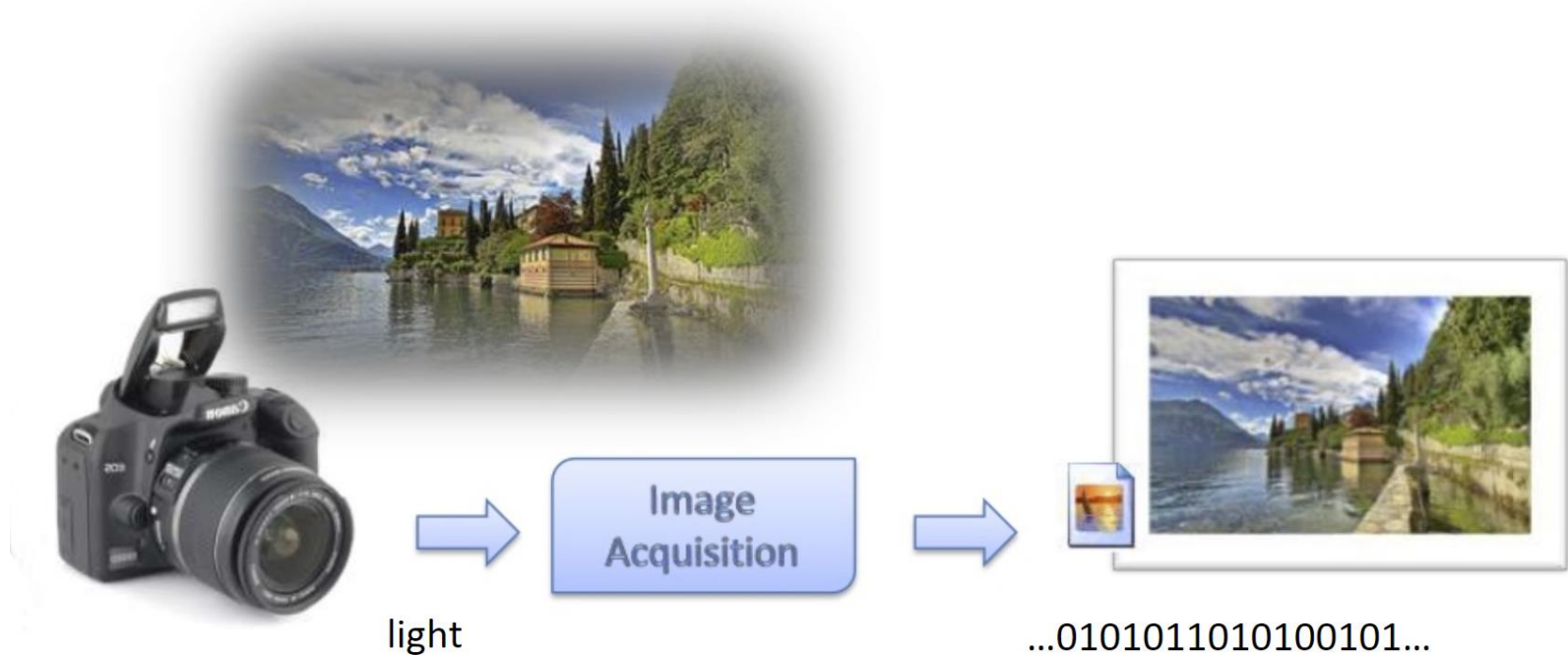
Usage of digital fingerprints

- For **source identification**:
 - Fingerprints are usually extracted and then compared with a dataset of possible fingerprints specific for each class/brand/model of acquisition (creation) devices
- For **forgery detection**:
 - detect non-uniformity or absence of fingerprints within the analyzed data
 - detect the presence of fingerprints pointing to a specific post-processing

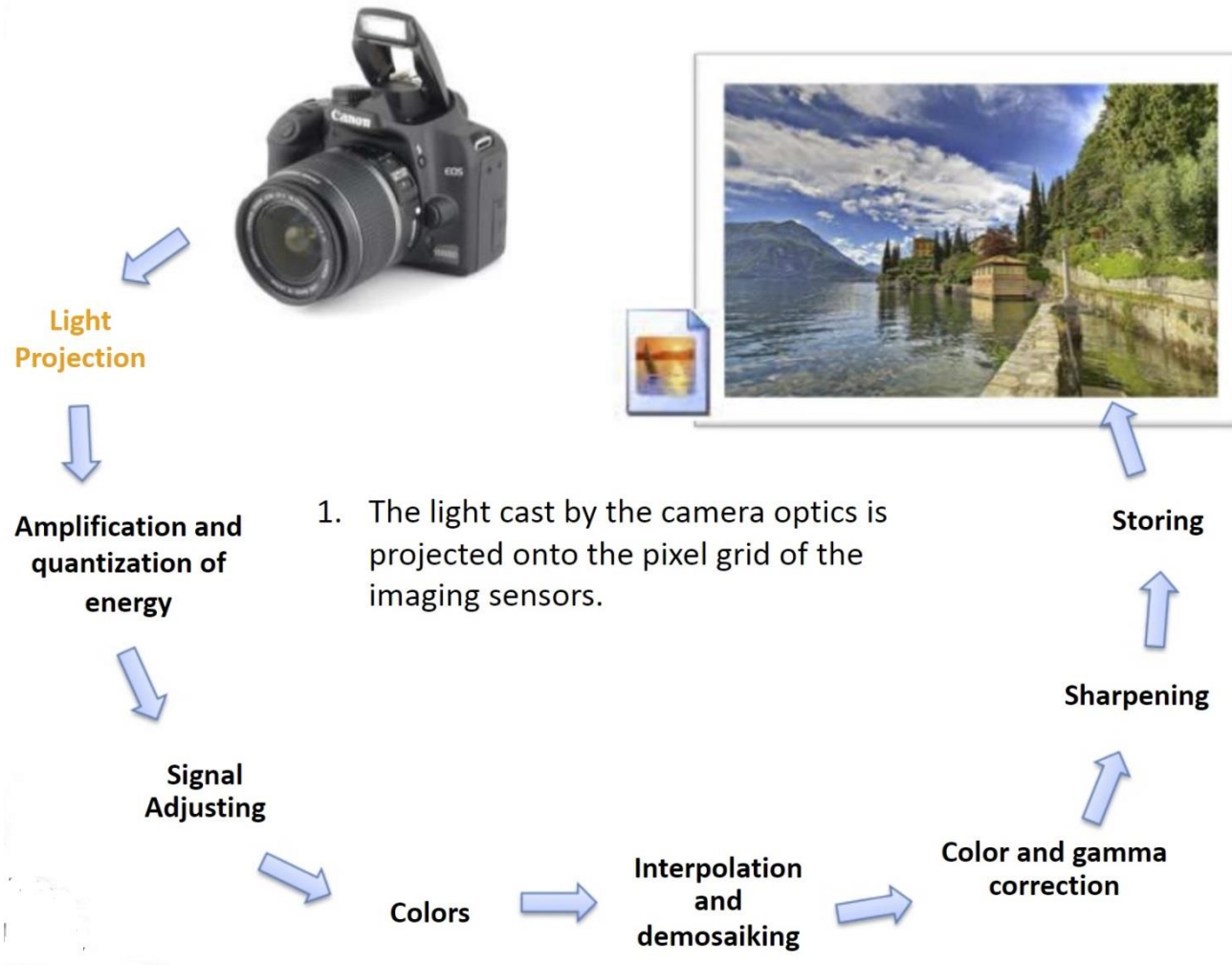
General principles of multimedia forensics

Image acquisition

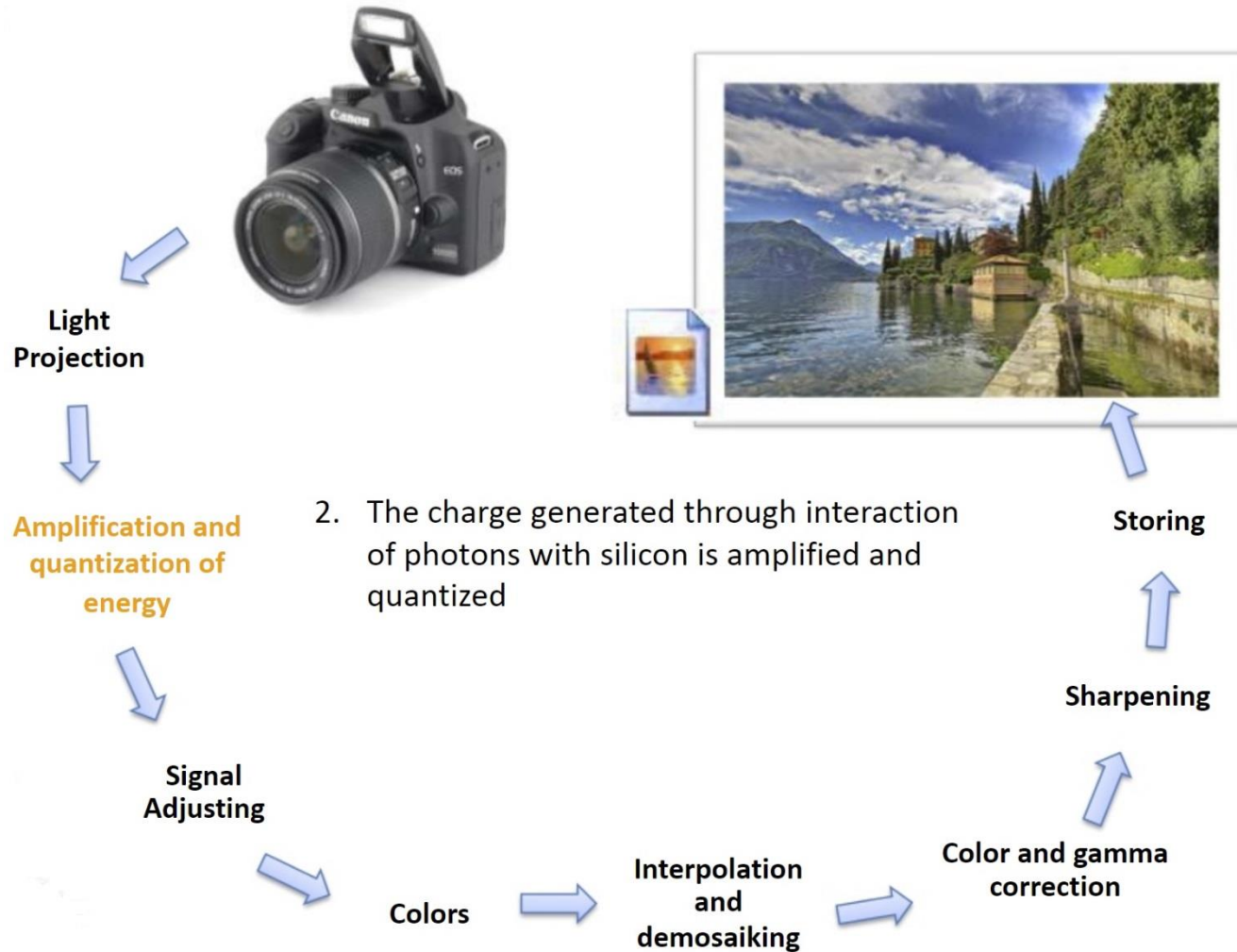
The process of acquiring a digital image is quite complex and varies significantly across different camera models, however, some basic elements are common for most cameras



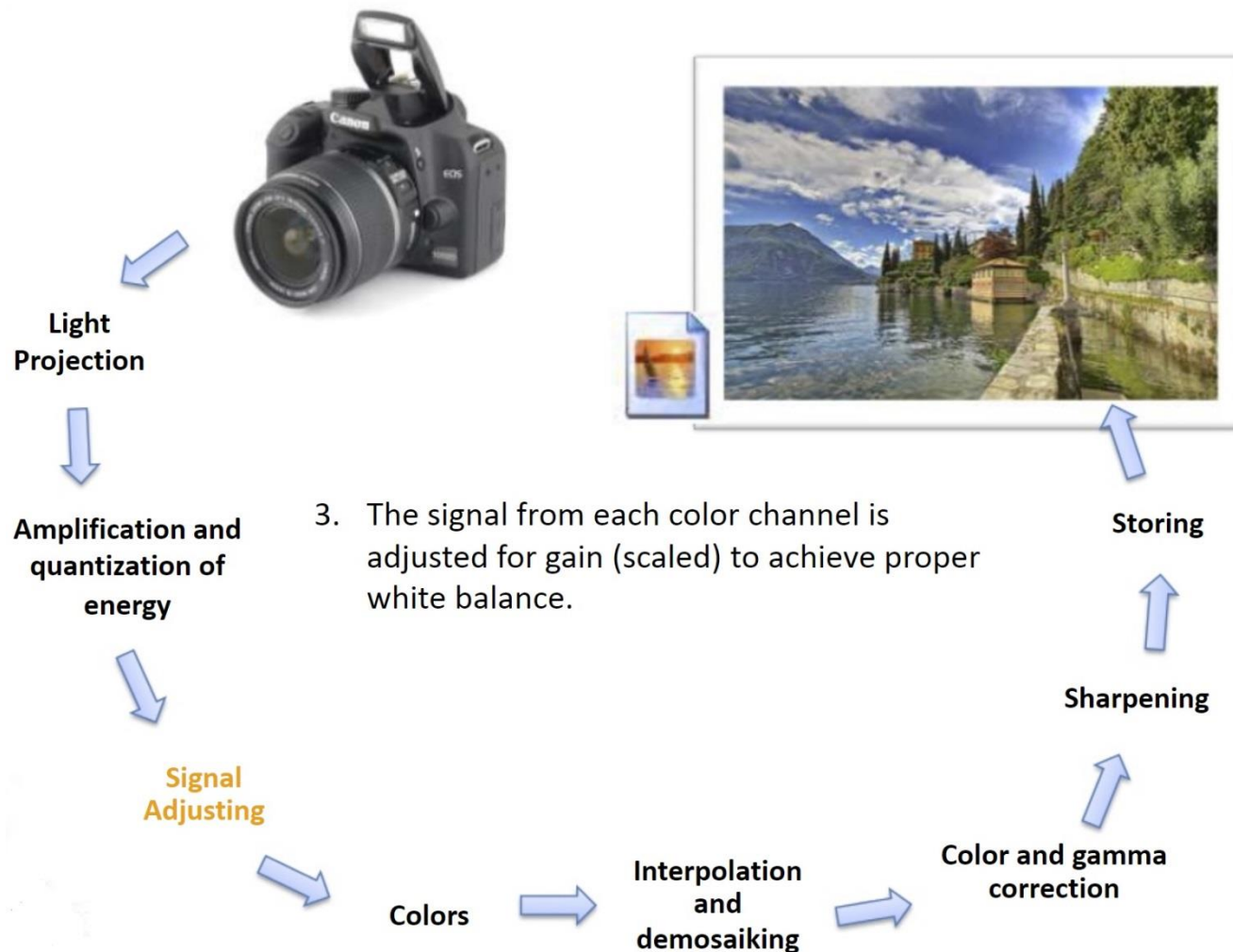
General principles of multimedia forensics



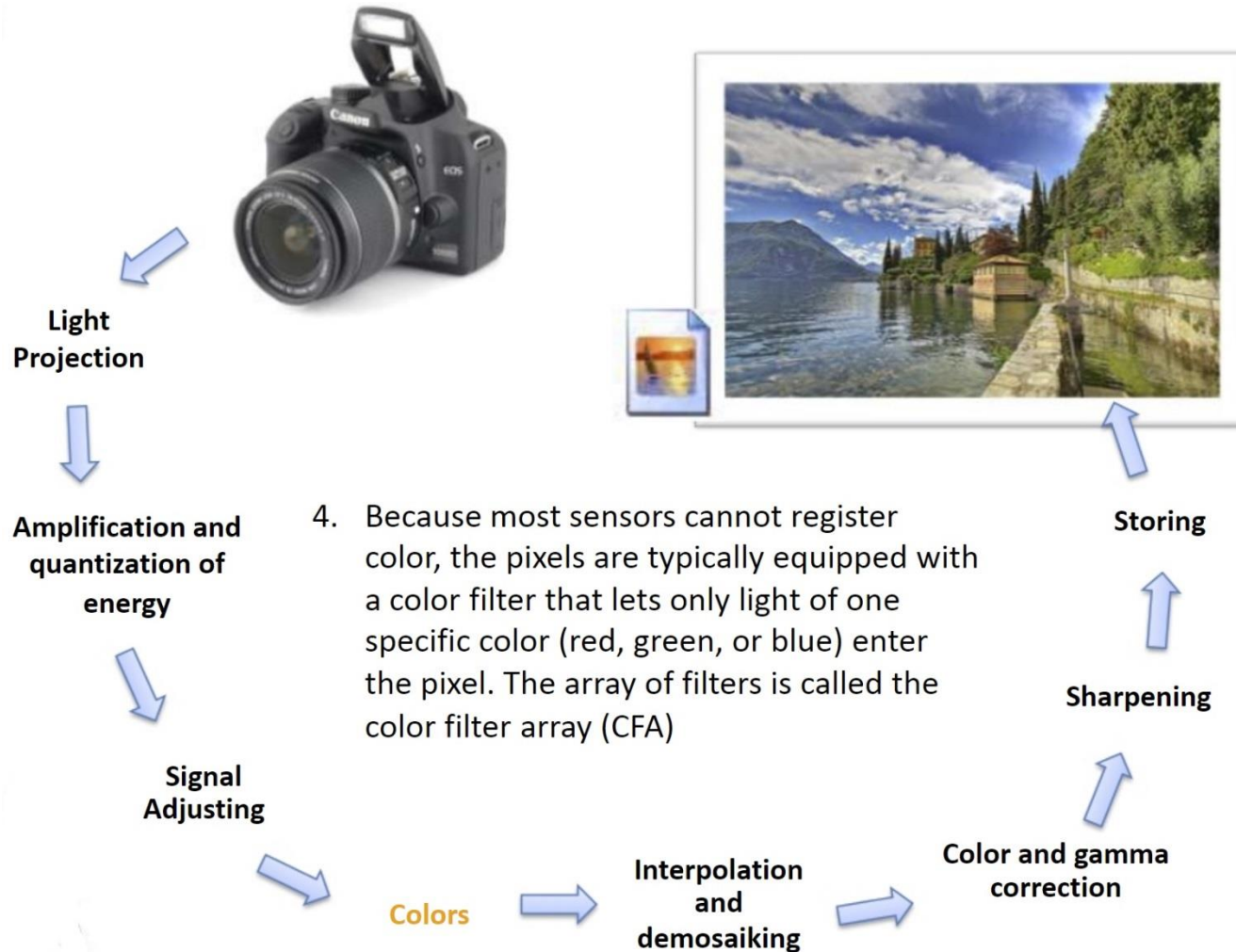
General principles of multimedia forensics



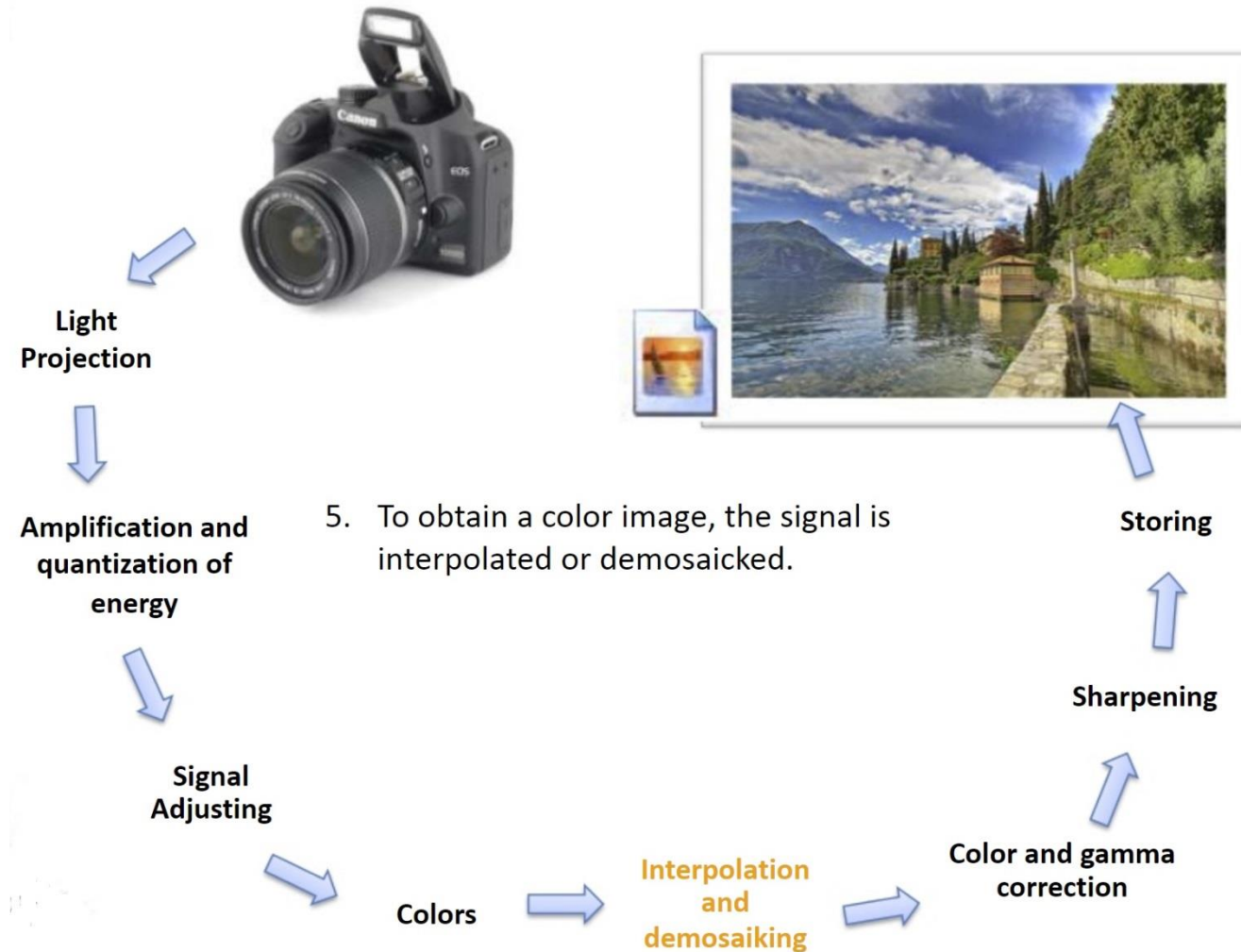
General principles of multimedia forensics



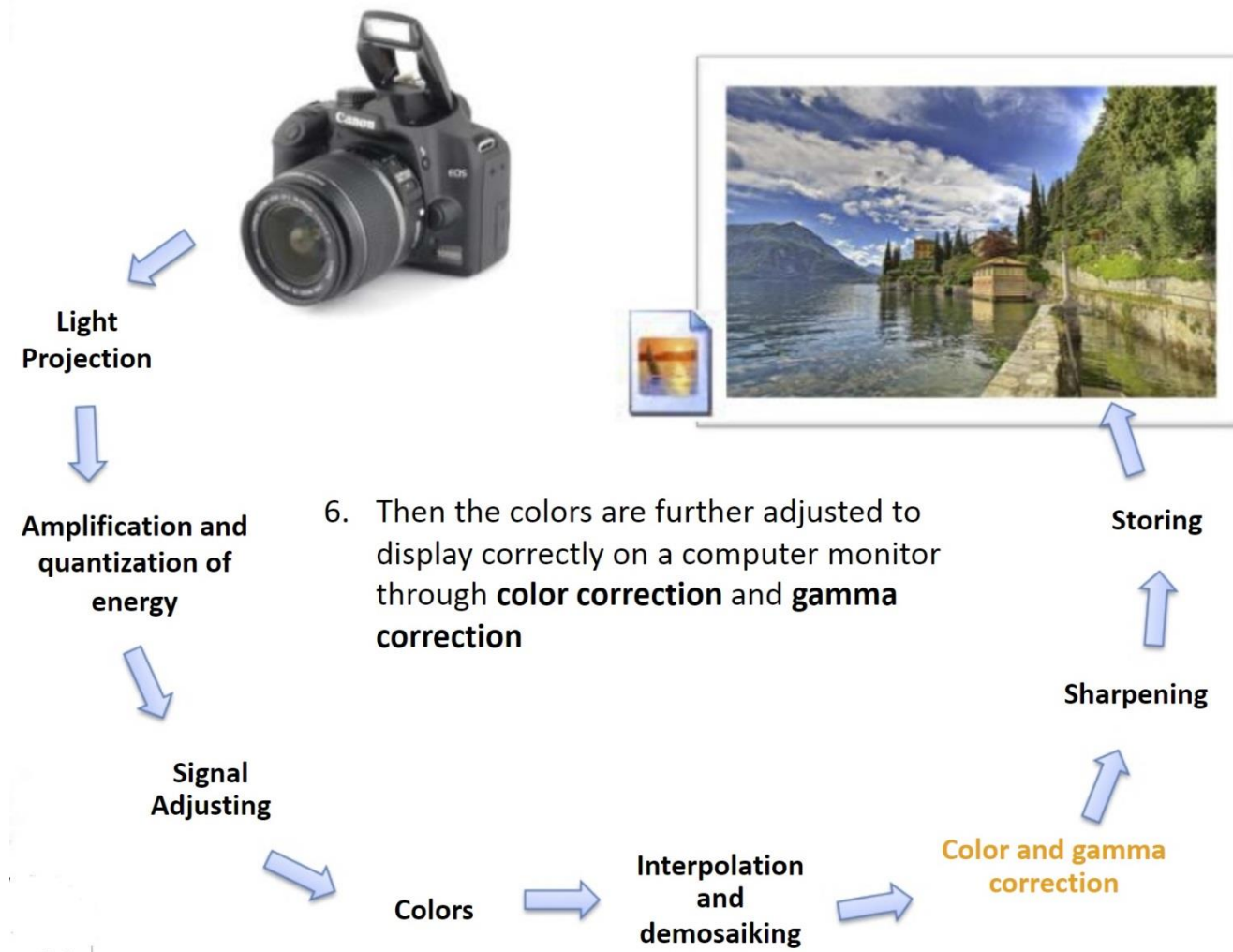
General principles of multimedia forensics



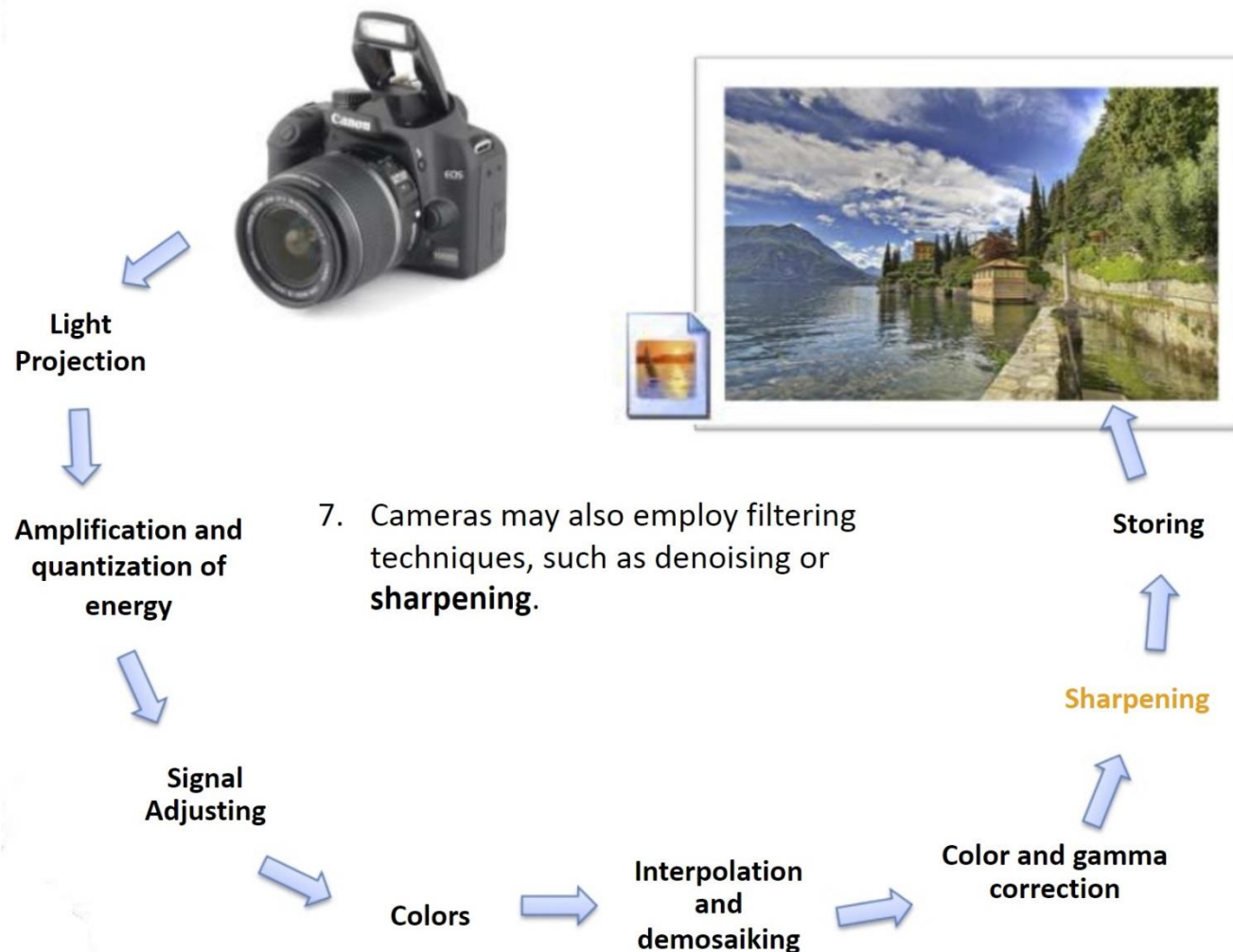
General principles of multimedia forensics



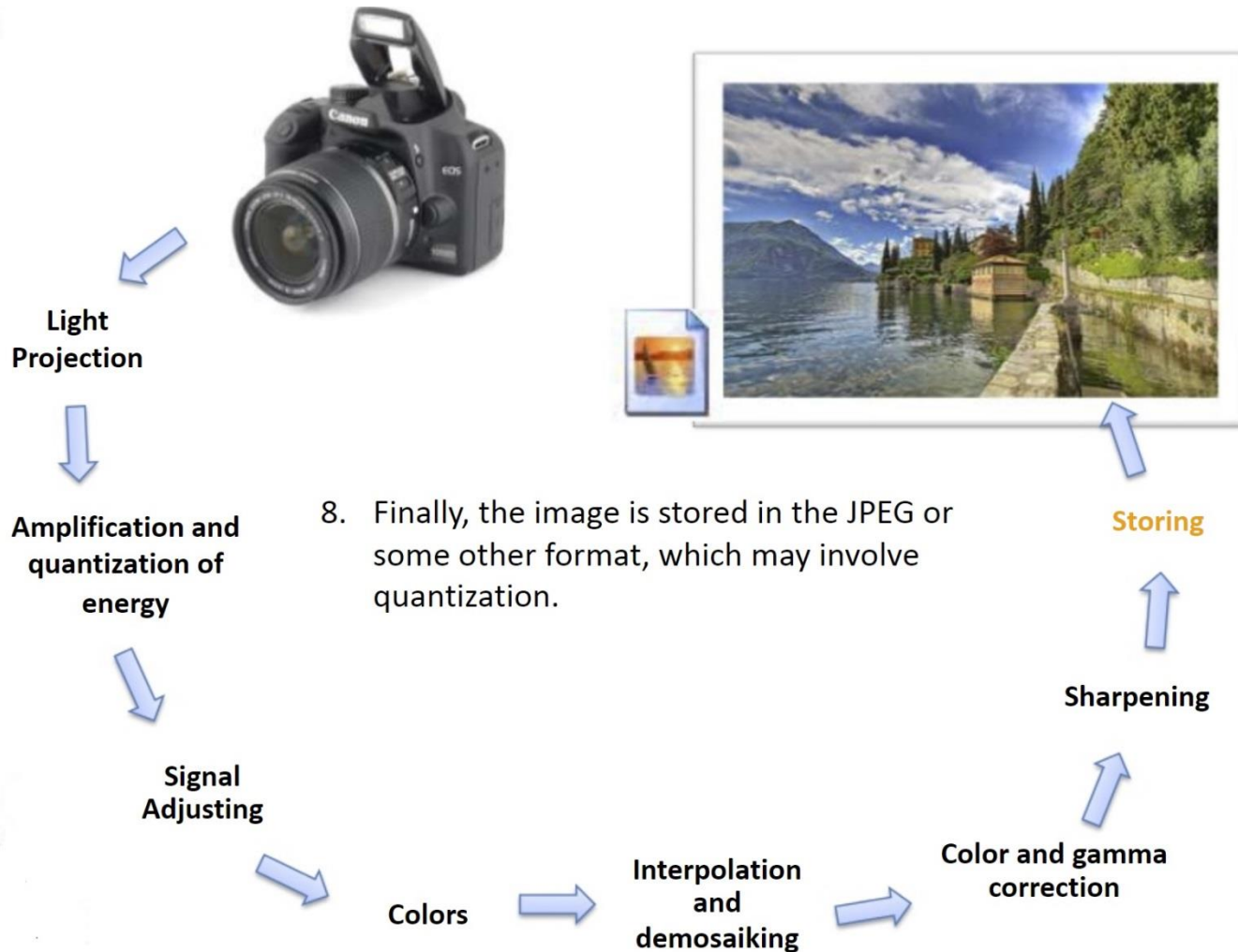
General principles of multimedia forensics



General principles of multimedia forensics

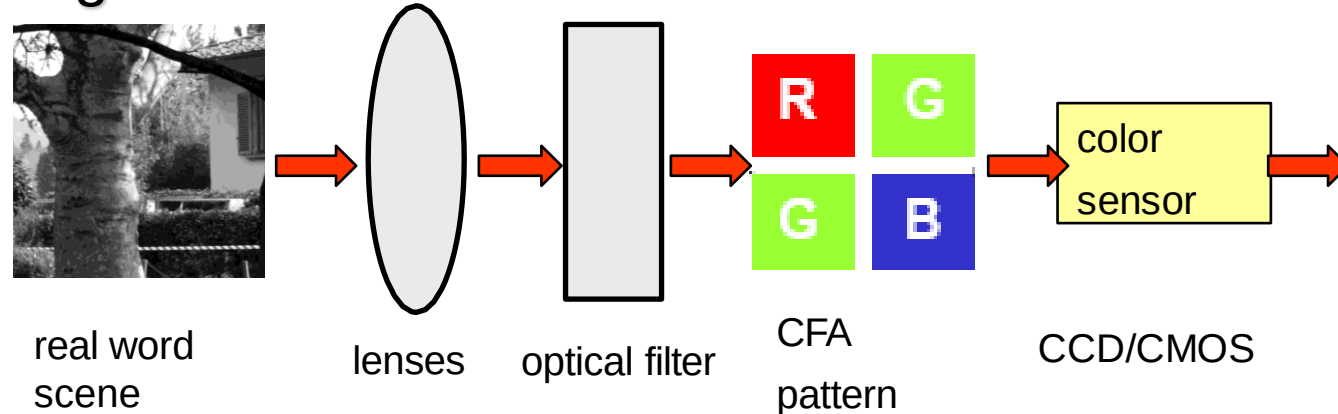


General principles of multimedia forensics

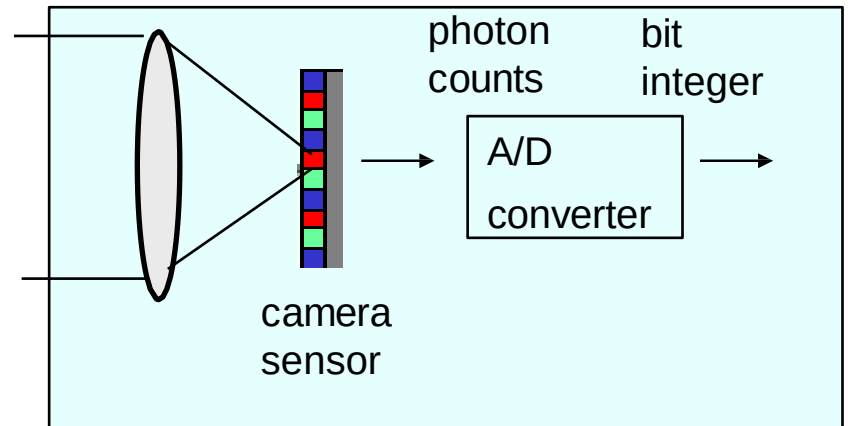


General principles of multimedia forensics

Digital Camera Model

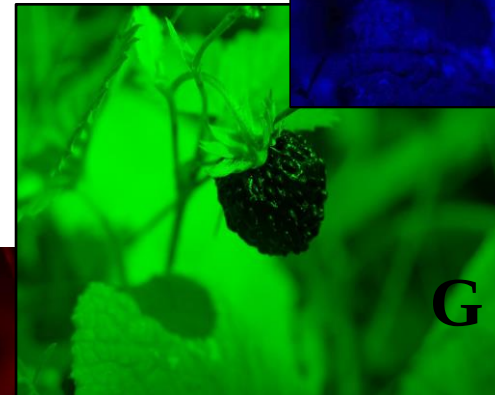


- Light is focused by the lenses on a 2D array of CCD/CMOS (pixels)
- Such elements are hit by the photons and convert them into voltage signals which are then sampled by an A/D converter
- Before reaching the sensor, the rays from the scene are filtered by the CFA (Colour Filter Array)



General principles of multimedia forensics

Bayer color array



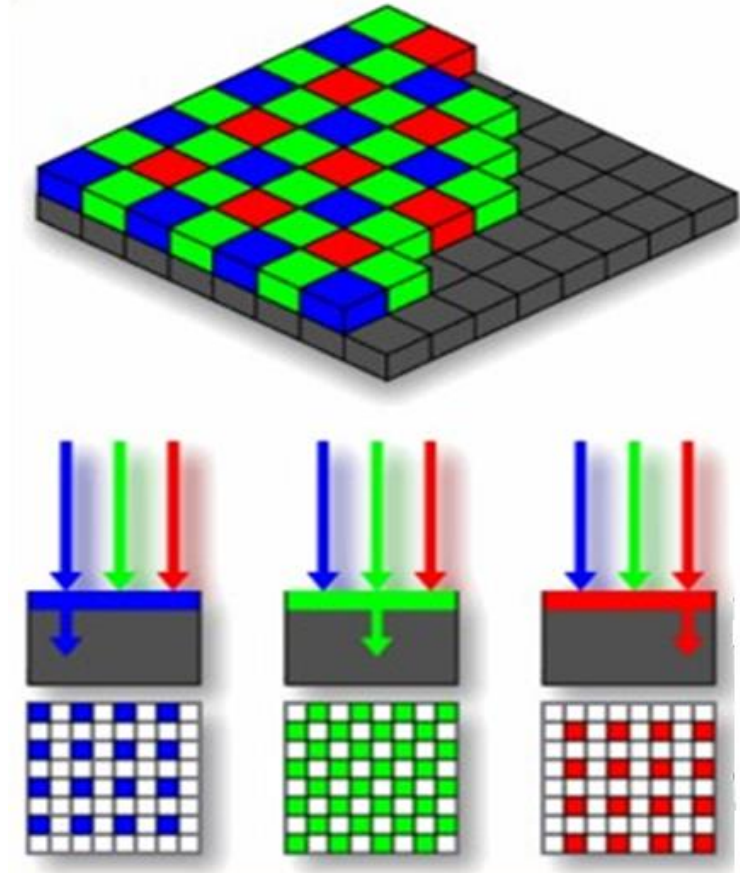
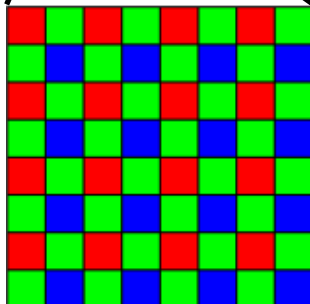
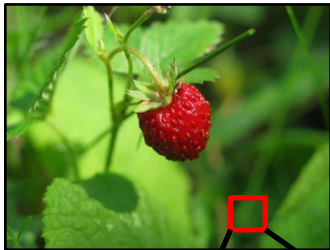
General principles of multimedia forensics

Bayer color array

Red pixels: 25%

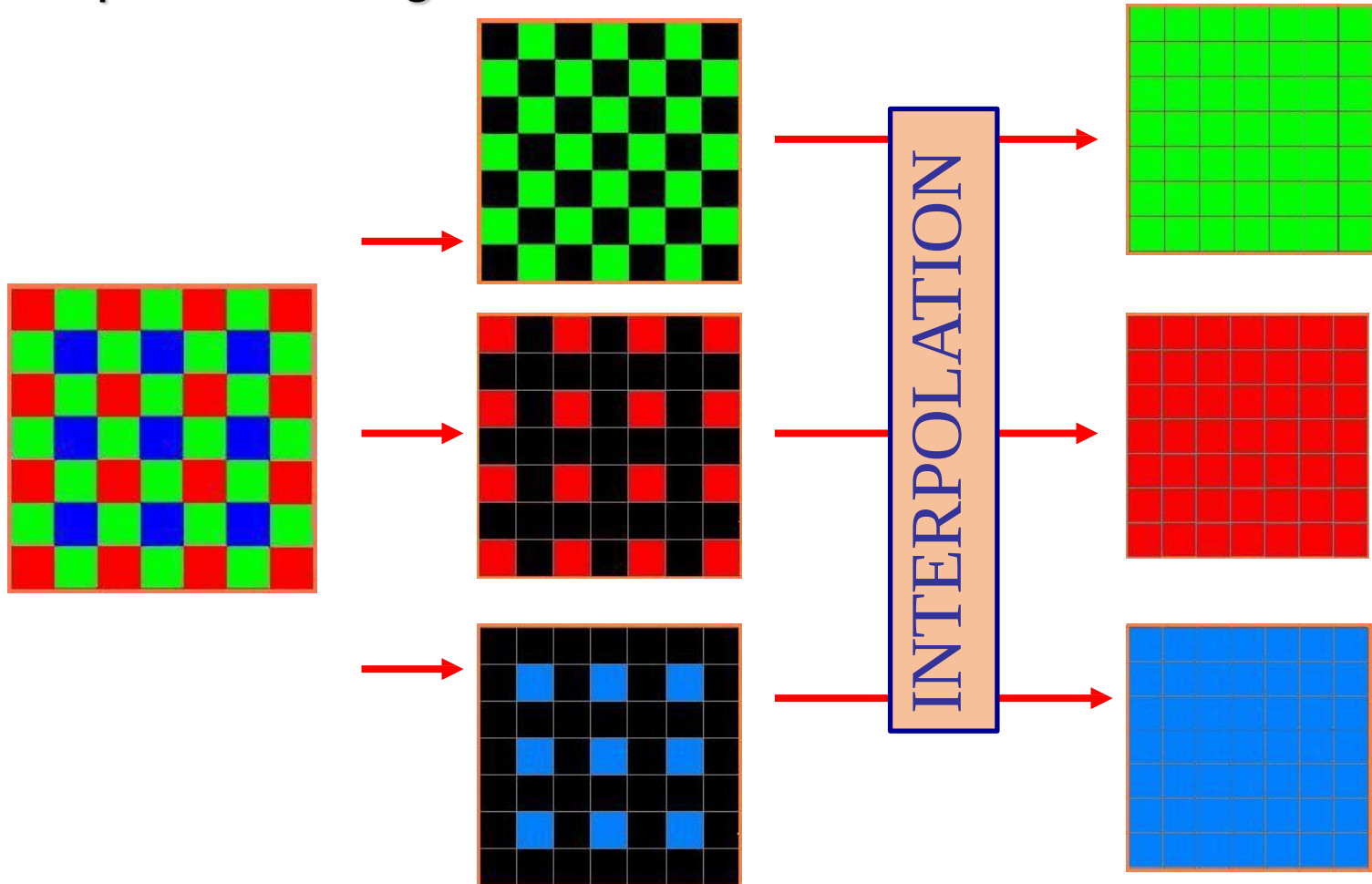
Green pixels: 50%

Blue pixels: 25%



General principles of multimedia forensics

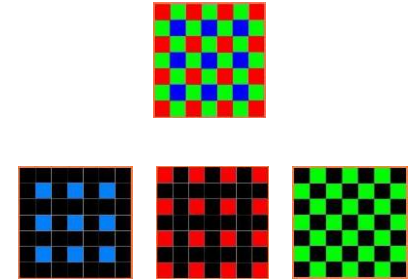
Color plain forming



General principles of multimedia forensics

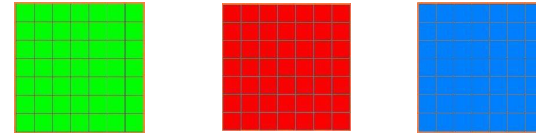
Source identification

- Bayer Array for almost all digital cameras



- Color interpolation algorithms are different for each digital camera manufacturer

Interpolation



- In the same way one can distinguish between different devices: scanners, computer generated images

General principles of multimedia forensics

Tampering detection



Incongruencies in CFA fingerprint can be used to detect tampering

General principles of multimedia forensics

Type of sensors

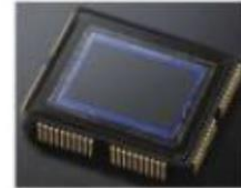
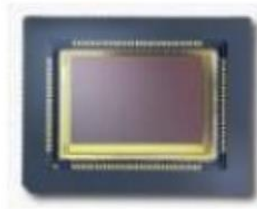
Image sensors are devices that convert light in electrical signals which will be later transformed in the corresponding bits of digital image

Types of digital camera sensors:

- **CCD (charge-coupled devices)**: currently the most commonly used image sensor, CCDs capture light onto an array of light-sensitive diodes, each diode representing one pixel
- **CMOS (complementary metal oxide semiconductor)**: made based on technology widely used to manufacture modern CPUs and memory chips, CMOS sensors are characterized by high noise immunity and low static power consumption

General principles of multimedia forensics

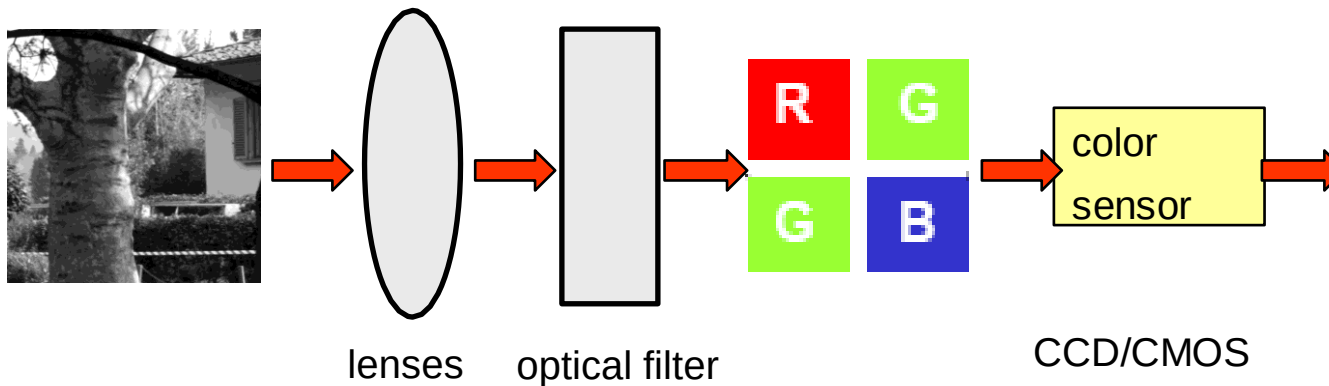
CCD vs CMOS



	CMOS	CCD
Cost	inexpensive because CMOS wafers are used for many different types of semiconductors	expensive to produce because of special manufacturing methods employed
Power	low power consumption	consumes upto 100x more power than CMOS
Noise	susceptible to noise	high quality, low noise images
Extended Functionality	other circuitry easily incorporated on same chip	technically feasible; other chips are used
Fill Factor	high	low

General principles of multimedia forensics

Sensors imperfections: noise



Sensor noise has 2 main components:

- Fixed Pattern Noise (FPN)
 - pixel to pixel difference in dark conditions
 - additive noise
- Photo-Response Non-Uniformity (PRNU)
 - dominant part of the pattern noise: multiplicative noise

General principles of multimedia forensics

Sensors imperfections: FPN

- The FPN is the pixel to pixel difference when the sensor is not exposed to light
- In most digital cameras this difference is equalized by subtracting a dark frame (mask) from the picture

The diagram illustrates the process of removing Fixed Pattern Noise (FPN) from a digital image. It shows a subtraction operation where a dark frame (mask) is subtracted from a noisy image to produce a clean image.

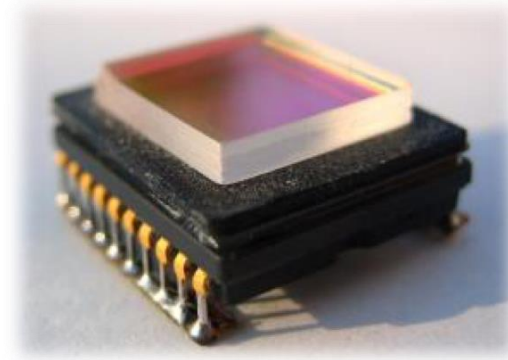
$$\begin{array}{cccc} \text{Noisy Image} & - & \text{Dark Frame (Mask)} & = & \text{Clean Image} \\ \begin{array}{|c|c|c|c|} \hline -2 & 0 & 3 & -5 \\ \hline \end{array} & - & \begin{array}{|c|c|c|c|} \hline -2 & 0 & 3 & -5 \\ \hline \end{array} & = & \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \end{array}$$

The visual representation shows two rows of four boxes each. The top row represents the 'Noisy Image' with values -2, 0, 3, and -5. The bottom row represents the 'Dark Frame (Mask)' with the same values. A minus sign is placed between the two rows. To the right of the rows is an equals sign, followed by a single row of four boxes representing the 'Clean Image' with values 0, 0, 0, and 0. The boxes in the original image have varying shades of gray corresponding to their values.

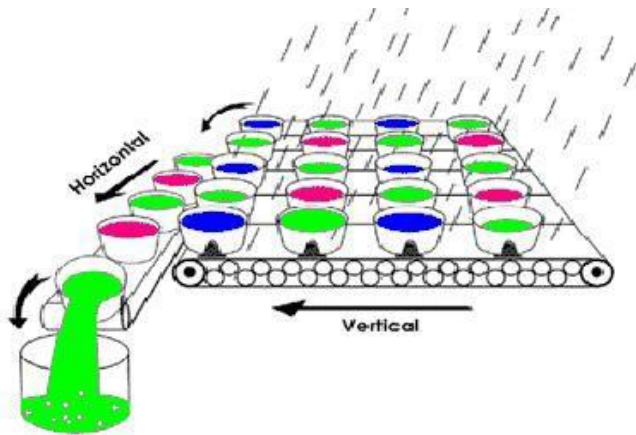
General principles of multimedia forensics

Sensors imperfections: PRNU

Typically, a digital camera has a 2D array of **several million elements**, each of which is responsible of the acquisition of a single pixel



A sensor is often exemplified as a **bucket collecting rain** (photons) until a certain level (the pixel value) is reached



Ideally, when uniform light falls on a camera sensor, each pixel should output exactly the same value

Practically, small variations in cell size and substrate material result in slightly different output values

General principles of multimedia forensics

Photo Response Non Uniformity

- Given an image I of size $M \times N$, image imperfections can be modeled as (*simplified model*):

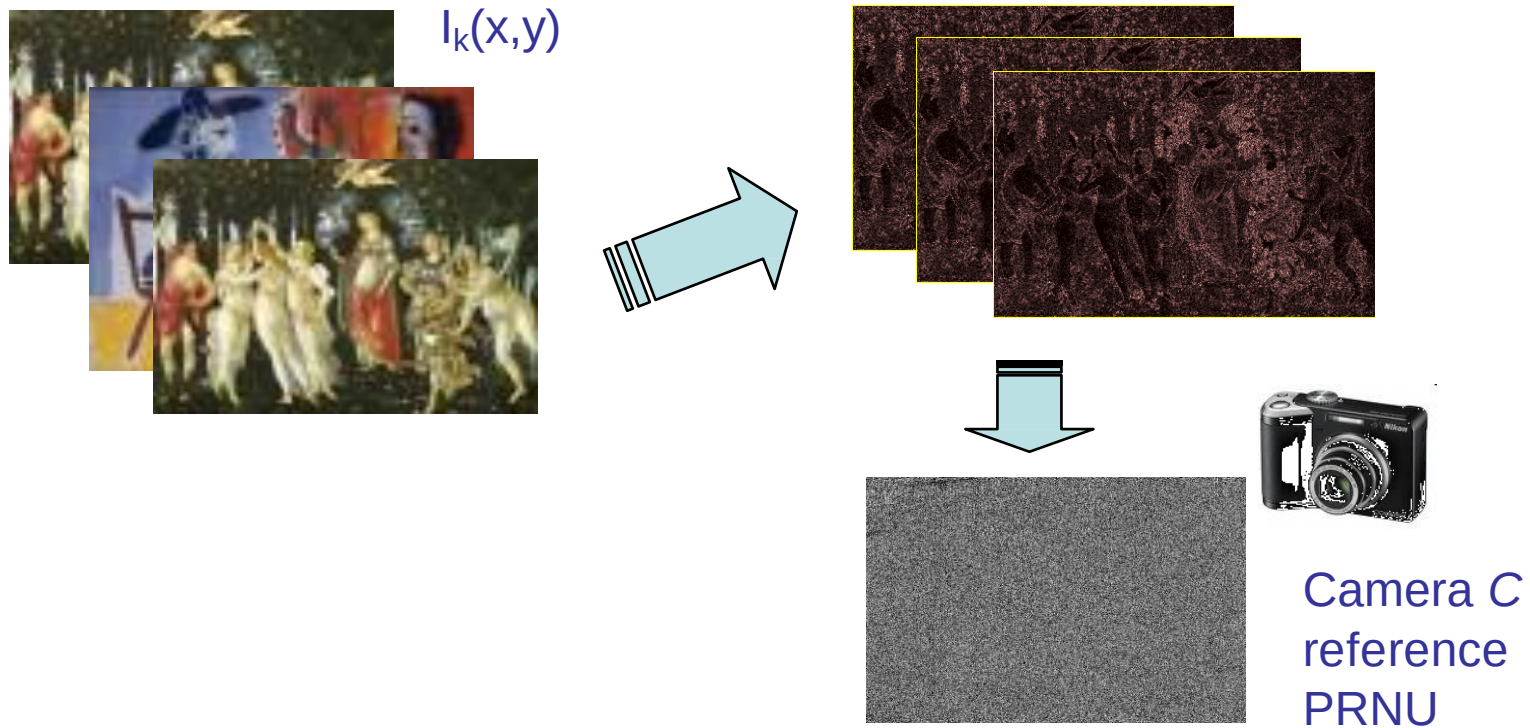
$$I(x,y) = I_0(x,y) + I_0(x,y) K(x,y) + N(x,y)$$

- $I_0(.)$ is the noise-free image, $K(.)$ multiplicative noise term (PRNU), $N(.)$ is an additive noise term (other disturbs).
- Goal: extract signal of interest $K(.)$ from observed data $I(.)$ -> **use of denoising tools**

General principles of multimedia forensics

Photo Response Non Uniformity (PRNU)

Estimation of PRNU for camera C is obtained by averaging noise residuals of a number of training images taken by camera C



General principles of multimedia forensics

Image residual: standard filters

$$W_I = I - F(I)$$

- **Gaussian smoothing, 2D-Wiener ...**
- **Advantages**
 - Simple implementation
 - Very fast
- **Disadvantages**
 - Image content left behind in the pattern **alters the correlation** between reference PRNU and the residual of the image under analysis

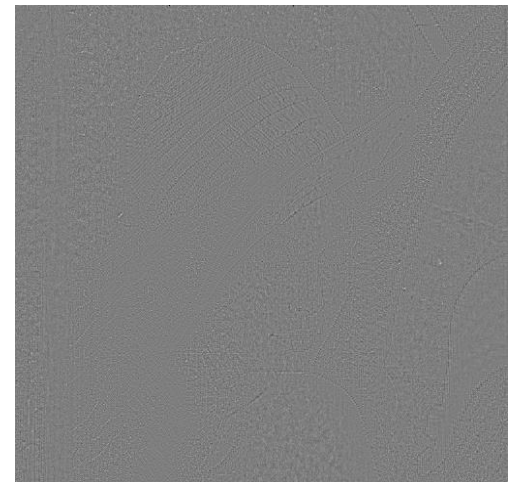


General principles of multimedia forensics

Image residual: best filter

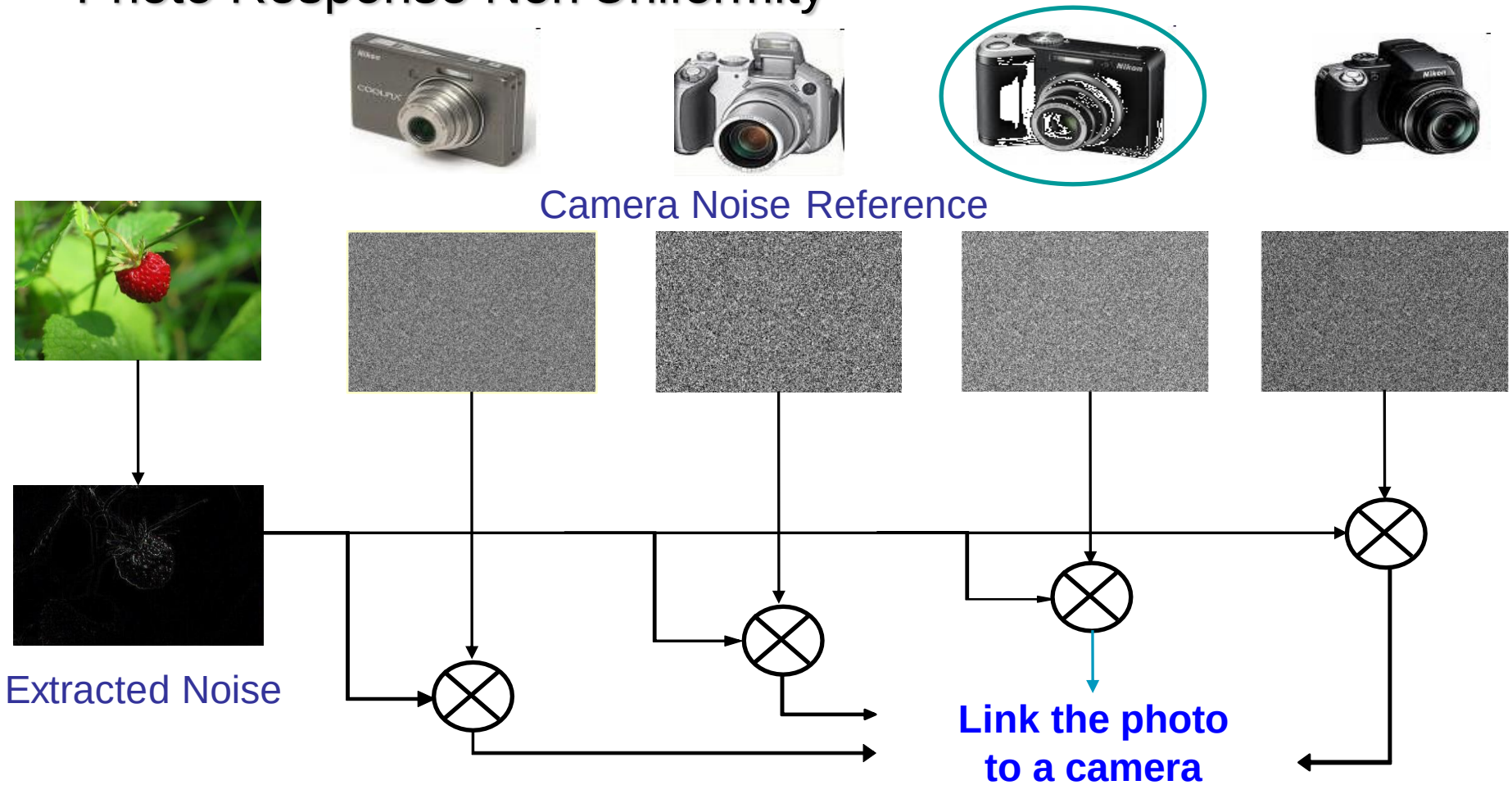
$$W_I = I - F(I)$$

- **Wavelet based denoising**
- **Advantages**
 - Significantly more accurate
 - Better PRNU fingerprint estimation
- **Disadvantages**
 - Slower
 - Higher complexity



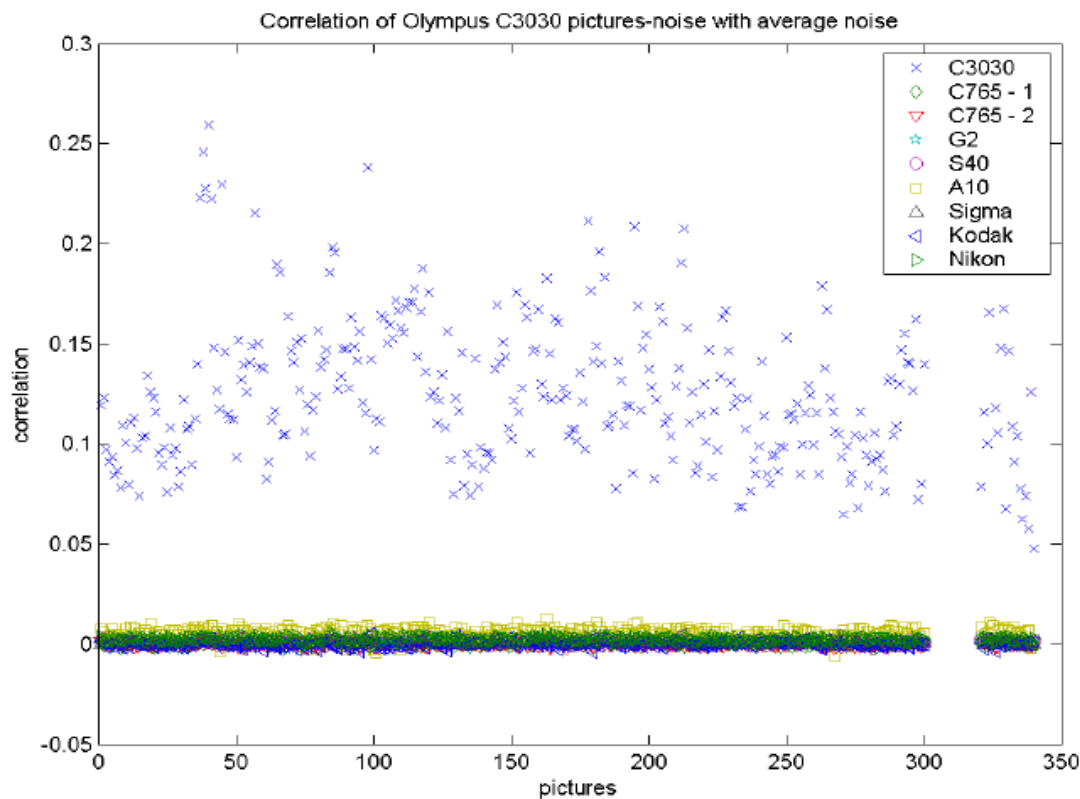
General principles of multimedia forensics

Photo Response Non Uniformity



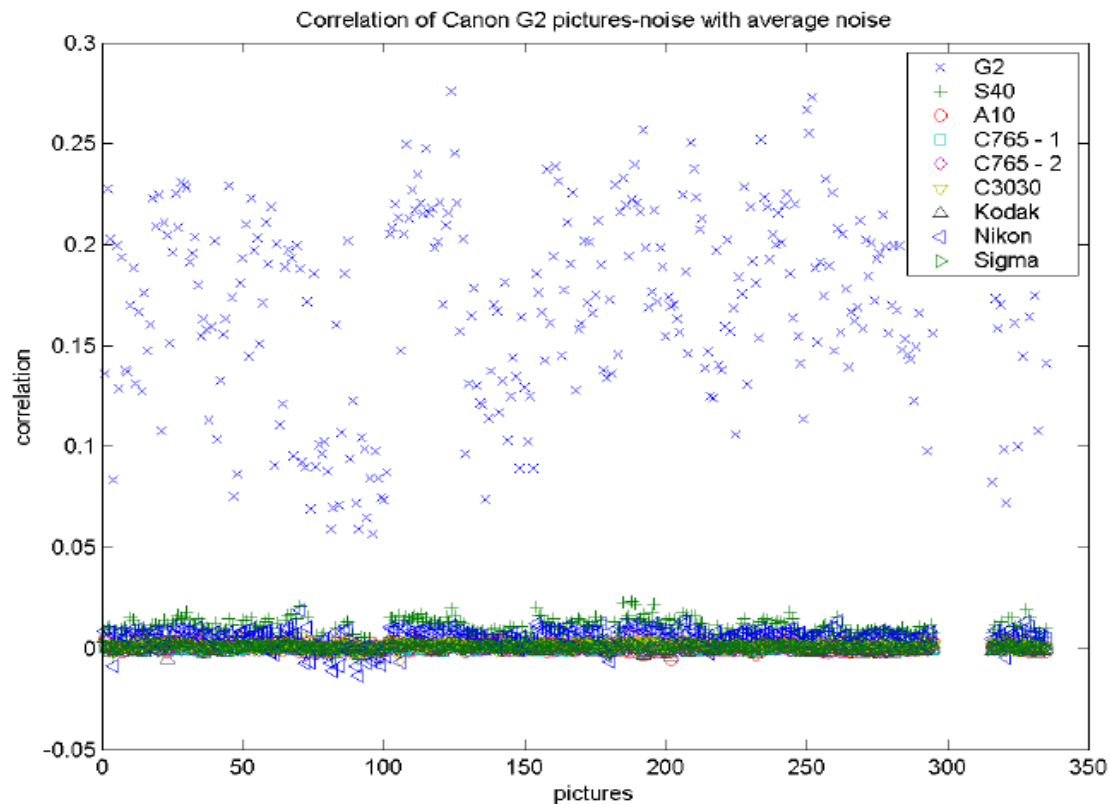
General principles of multimedia forensics

Olympus 3030 (all JPEGs)



General principles of multimedia forensics

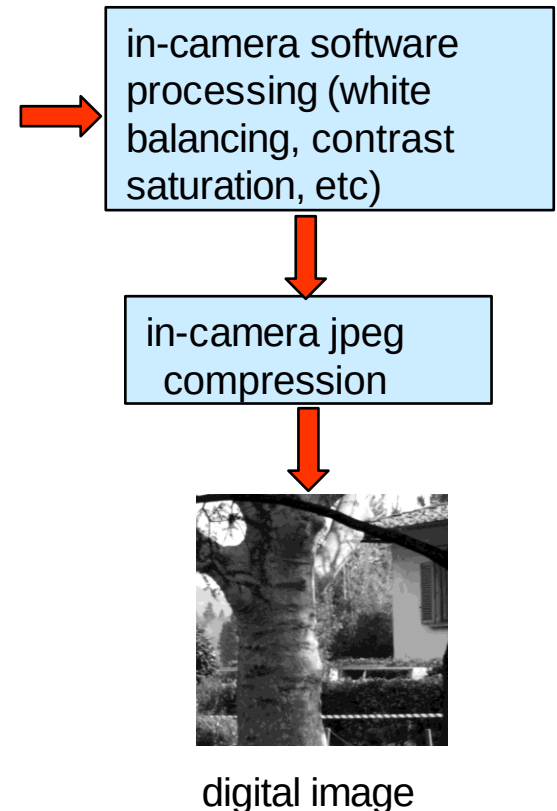
Canon G2 (raw)



General principles of multimedia forensics

Digital Camera Models

- The signal undergoes additional processing such as: white balancing, color processing, image sharpening, contrast enhancement, gamma correction
- It is stored in the camera memory in a customized format, (for commercial devices JPEG format is usually preferred)
- All these steps introduce traces that can be exploited for multimedia forensics analysis



General principles of multimedia forensics

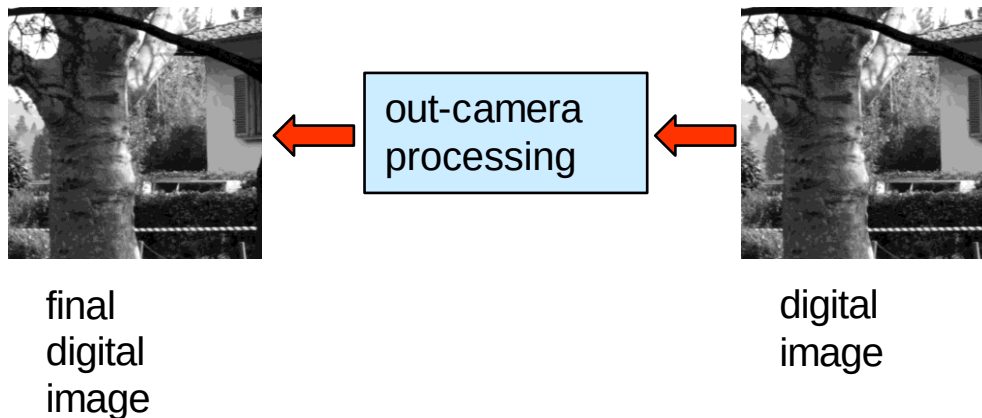
Digital Camera Models



General principles of multimedia forensics

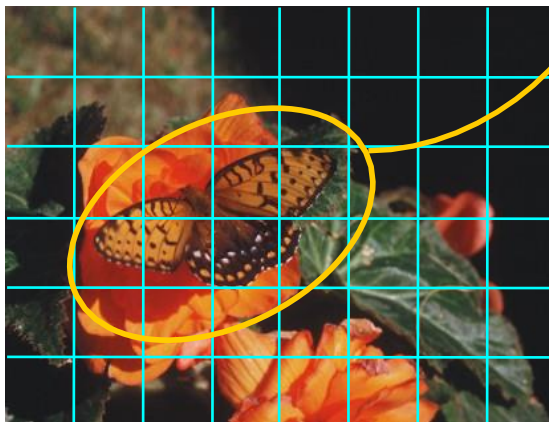
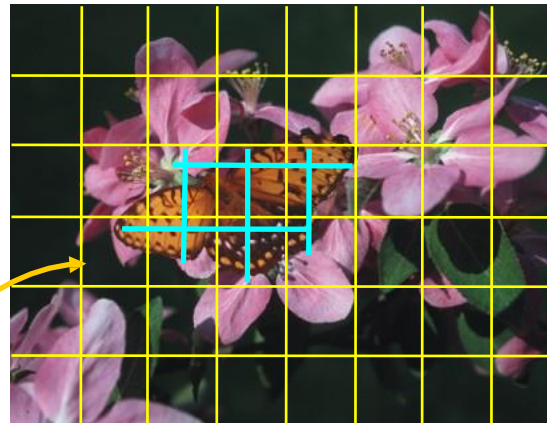
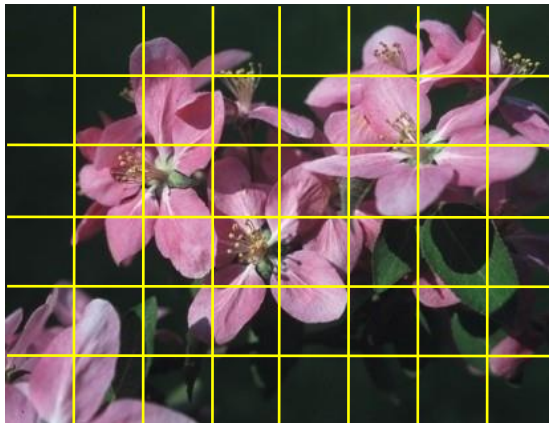
Out-camera processing

- Several kinds of processing can be applied to an image during its lifecycle:
 - compression
 - geometric transformation (rotation, scaling, ...)
 - blurring and sharpening
 - contrast adjustment
 - etc.



Examples of media forensic applications

Double JPEG artifacts



- JPEG compression leaves artifacts at the border of 8x8 blocks
- in case of double compression, the traces of old and new compression stages are likely to be desynchronized, providing room for multimedia forensic analysis

Examples of media forensic applications

Geometric fingerprint: copy move

This is a particular kind of tampering: part of image is duplicated to cover some undesired details



Examples of media forensic applications

Geometric fingerprint: copy move

Possible solution: we analyze the image with a sliding window looking for improbable duplicates



However, it is not possible to understand which of the duplicated parts is the original

Examples of media forensic applications

Geometric fingerprint: highlights



The eyes are modeled as spheres and infer the direction of light source from highlights

Very likely this picture was taken in three different time instants

Examples of media forensic applications

Geometric fingerprint: shadows



Under certain assumptions
it is possible to derive the
direction of light from
shadows



Examples of media forensic applications

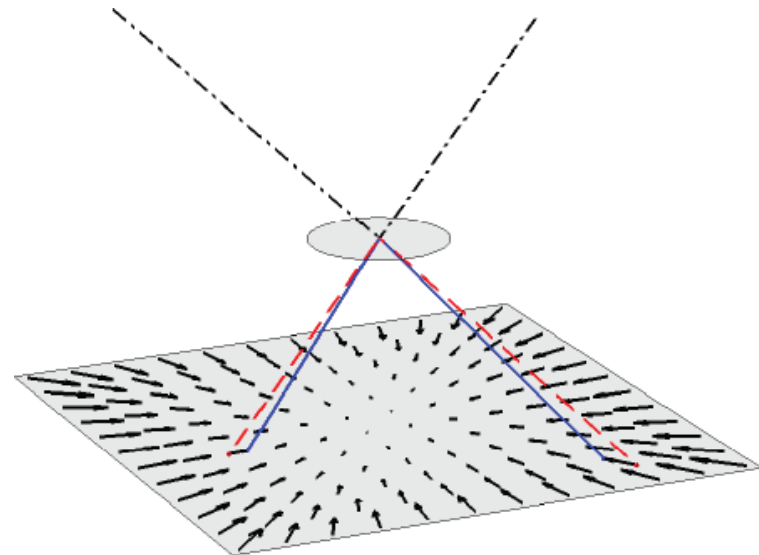
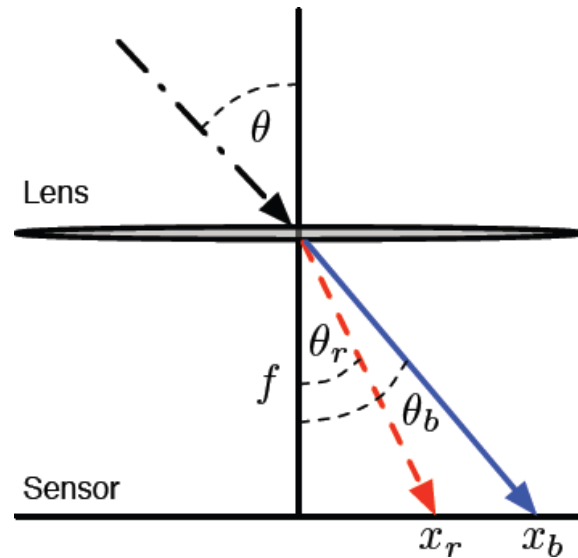
Geometric fingerprint: shadows



Creating a photomontage by preserving the coherence of light and shadows is not an easy task

Examples of media forensic applications

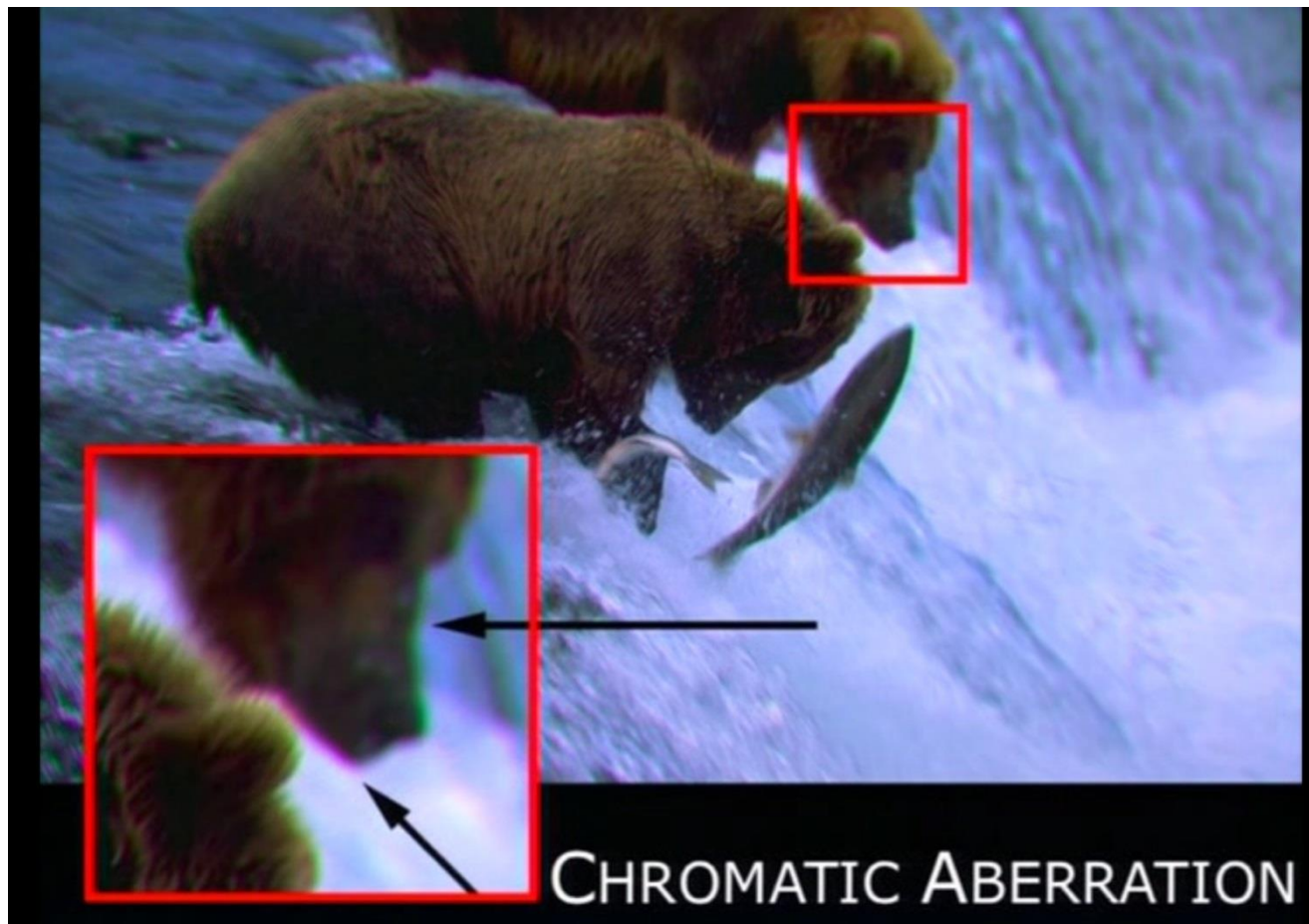
Geometric fingerprint: chromatic aberration



Non-planar lenses create a rainbow effect due to light aberration, which can be used to detect cut & paste tampering

Examples of media forensic applications

Geometric fingerprint: chromatic aberration



Examples of media forensic applications

AI from threat to defense

- AI (deep learning) capabilities can be exploited to identify image source, detect processing operations, tampering detection
- Significant advances made in last 20 years by applying CNN architectures for forensics
- Race of arms between AI and AI
- Vulnerability to intentional informed attacks

Conclusions and references

- Multimedia forensics is relatively young yet active area of research
- A Road Map for Digital Forensic Research (First Forensic Research Workshop, 2001):
https://www.dfrws.org/sites/default/files/session-files/a_road_map_for_digital_forensic_research.pdf
- Hany Farid (UC Berkeley, USA):
<https://www.ischool.berkeley.edu/people/hany-farid>
- Jessica Fridrich (Binghamton University, USA):
<http://www.ws.binghamton.edu/fridrich>
- Min Wu (University of Maryland, USA):
<https://user.eng.umd.edu/~minwu>
- Mauro Barni (University of Siena, Italy):
<https://scholar.google.it/citations?user=ntRScY8AAAAJ&hl=en>
- Nasir Memon (New York University, USA):
<https://engineering.nyu.edu/faculty/nasir-memon>